Credit Markets, Uncertainty, and Business Cycles*

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Abstract

This paper studies the sources of cyclical fluctuations in U.S. macroeconomic activity. Hours worked has a pronounced spectral peak at business-cycle frequencies, yet the shock that explains most of its variation does not display a clear boom–bust response. Using a Bayesian vector autoregression, we show that this dominant movement combines two distinct shocks. An increase in macro uncertainty contracts output, consumption, investment, and hours, after which activity gradually returns toward steady state. A separate credit-spread shock, orthogonal to the uncertainty shock, produces an expansion followed by a contraction. This second shock also generates a boom–bust response in stock prices and accounts for substantial variation in the excess bond premium, linking it to credit-market sentiment. We develop a model in which firms borrow against collateral to finance their wage bill. Anticipated productivity news relaxes financing conditions, while the non-realization of that news reverses the expansion. Uncertainty about enforcement instead produces a contractionary, mean-reverting response. Nominal wage rigidity allows both shocks to generate comovement in the main macroeconomic aggregates.

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1 Introduction

A common narrative in macroeconomics is that recessions are preceded by expansions: economic activity rises for a time, loses momentum, and eventually contracts. In U.S. data, we find patterns consistent with this narrative. The spectral density of hours worked is concentrated within the range of periodicities conventionally associated with business cycles and has a pronounced local peak, while its autocorrelation eventually becomes negative. For a single shock, these facts require impulse responses of opposite signs at different horizons. Yet the shock that accounts for most of the business-cycle variation in hours does not display a clear expansion followed by a contraction. Output, consumption, investment, and hours rise together and then return gradually toward baseline. Where is the cycle indicated by spectrum but largely absent from the impulse response?

This paper argues that the answer lies in the composition of the movement that explains most of the variation in hours (the Main Business Cycle shock, following Angeletos et al. (2020)). The Main Business Cycle (MBC) shock combines two disturbances with different propagation. An increase in macro uncertainty causes an immediate recession: output, consumption, investment, and hours fall together and subsequently recover. A separate credit-market shock produces the missing cycle. Easier credit conditions generate a broad expansion, but the expansion is followed by a contraction in the all four macro variables. Within the MBC shock, the larger uncertainty component obscures the reversal associated with credit markets. Separating the two disturbances reconciles the frequency-domain evidence with the time-domain responses.

We establish these results in three steps. We begin by documenting the frequency- and time-domain properties of hours worked over the past half century, from 1973:I through 2024:IV. Its spectral density peaks at extended business-cycle periodicities of 6 to 40 quarters, while the autocorrelation of filtered hours becomes negative at longer horizons. We then estimate an eight-variable Bayesian VAR that jointly tracks aggregate activity, monetary conditions, productivity, credit spreads, and macro uncertainty. Following the max-share approach of Uhlig (2003), we identify the MBC shock as the shock that accounts for the largest share of fluctuations in hours within the 6-to-40-quarter band. It explains about three quarters of the fluctuations in hours within this band, more than 60 percent of the fluctuations in output and investment, and roughly half of those in consumption. It also accounts for a substantial part of movements in credit spreads and macro uncertainty. Yet, although the hours spectrum attributable to the MBC shock has a pronounced peak, its impulse response displays no clear boom–bust pattern.

To understand this contrast, we next turn to the financial variables. A shock that jointly maximizes the business-cycle variation in macro uncertainty and the credit spread generates responses that are nearly identical to those of the MBC shock, and the two shock series have a correlation of 0.97. The dominant movement in hours is therefore closely associated with the common movement in uncertainty and credit spreads. This correspondence, however, need not mean that a single structural innovation drives both variables. Because a max-share criterion

isolates their dominant common direction, it can combine disturbances that propagate differently over time.

The third step separates these forces using a sequential penalty-function approach related to Caldara et al. (2016). The first shock captures the largest possible movement in macro uncertainty. Higher uncertainty produces a broad recession: output, consumption, investment, and hours fall together and then gradually recover, without a subsequent expansion. This shock accounts for nearly four-fifths of the business-cycle variation in uncertainty. We then identify a second shock that is orthogonal to the uncertainty shock and captures the remaining joint movement in uncertainty and credit spreads. A decline in the credit spread generates an expansion lasting roughly three years, followed by a contraction that is especially clear in output, investment, and hours. This second shock explains more than half of the business-cycle variation in the credit spread but little of the variation in uncertainty, and its contribution to the hours spectrum is concentrated near the observed peak. Thus, the cycle hidden within the MBC shock comes primarily from credit markets rather than from uncertainty.

The behavior of other financial variables gives the second shock an economic interpretation. Stock prices rise during the expansion and fall below their starting point during the contraction. The excess bond premium initially falls and later rises, while the share of high-yield issuance first increases and then reverses. The shock also accounts for more than half of the variation in the excess bond premium. This combination of low risk compensation, greater risky issuance, high equity values, and a subsequent reversal resembles the credit-market sentiment dynamics documented by Greenwood and Hanson (2013) and López-Salido et al. (2017). We therefore interpret the second disturbance as a credit sentiment shock. This interpretation follows from its macroeconomic and financial effects; it is not imposed as part of the identification.

These findings place a sharp restriction on theory. A model must generate comovement in output, consumption, investment, and hours after both shocks, but the dynamics must depend on which shock occurs. The credit sentiment shock must produce an expansion followed by a contraction, whereas higher uncertainty must produce a contraction followed by a return to normal. Models in which every shock triggers oscillation cannot account for this distinction. A standard flexible-price response to uncertainty is also insufficient: precautionary saving lowers consumption but tends to raise labor supply, output, and investment (Basu and Bundick, 2017). The challenge is therefore to make the shape of the response depend on the source of the shock while preserving comovement among the four macroeconomic quantities.

We address this challenge with a rational-expectations model built around a common financing channel. Firms must pay workers before receiving their sales revenue and therefore borrow against the value of installed capital. Because equity payouts cannot move freely, changes in financial conditions alter the value of internal funds and, in turn, firms' demand for labor. Investment adjustment costs allow the value of installed capital, and hence borrowing capacity, to respond immediately even though the capital stock is predetermined at the time of the shock. Slow

adjustment of nominal wages then allows labor demand to move hours together with output, consumption, and investment.

The same financing channel produces different dynamics after the two shocks. Greater uncertainty about future borrowing capacity makes internal funds more valuable and discourages hiring and investment, generating a joint contraction followed by recovery. The credit cycle instead begins with favorable news about future productivity. Firms invest and borrow in anticipation of higher future productivity, collateral values rise, and financing conditions ease. If the news fails to materialize, the expected productivity gain disappears but the debt and investment inherited from the expansion remain. Collateral values fall relative to outstanding debt, borrowing conditions tighten again, and the expansion turns into a contraction. Credit-market optimism and pessimism therefore arise endogenously from anticipation and disappointment under rational expectations.

Because the borrowing constraint can switch between binding and slack, we solve the model globally using a Deep Equilibrium Network and estimate it by matching the empirical responses to the two shocks. The estimated model reproduces the central distinction in the data. Higher macro uncertainty generates a hump-shaped contraction in all four macroeconomic quantities. Favorable news that is later unfulfilled generates an expansion followed by a persistent contraction. Model-implied Tobin's Q also rises and then falls, paralleling the boom–bust response of stock prices in the data. The model does not match every feature equally well: it overstates parts of the investment and contractionary responses to unfulfilled news, and its consumption response is more persistent than the empirical median. These differences indicate where the mechanism can be improved without changing its central implication.

The rest of the paper proceeds as follows. The next subsection places the paper in the related literature. Section 2 describes the data and empirical method, documents the frequency- and time-domain properties of hours, and then separates the MBC shock into uncertainty and credit sentiment shocks. It concludes by drawing out the implications of these findings for theory. Section 3 develops the model's financing mechanism, introduces the two shocks, and explains how they generate different dynamics through the same channel. It then presents the calibration, estimation, and quantitative results. The final section concludes.

1.1 Literature Review

This paper relates first to work that uses a small number of empirical shocks to characterize business-cycle comovement. The max-share method of Uhlig (2003) identifies the innovation that accounts for the largest fraction of a variable's fluctuations. Angeletos et al. (2020) extend this idea across variables and frequency bands and show that shocks selected by the business-cycle variation in output, investment, hours, or unemployment leave similar macroeconomic footprints. We use the same frequency-domain logic, but our question is different. The shock that explains the largest share of hours at business-cycle frequencies produces a spectral peak without an

evident boom–bust impulse response. We show that this apparent discrepancy reflects the combination of two disturbances with different dynamics. To separate them, we adapt the penalty-function approach of Caldara et al. (2016), first identifying an uncertainty shock and then an orthogonal shock that maximizes movements in credit spreads. Thus, our empirical contribution is to connect the frequency-domain evidence of a cycle to distinct uncertainty and credit-market innovations in the time domain.

Our findings also contribute to the literature on credit conditions and future economic activity. Gilchrist and Zakrajšek (2012) show that the excess bond premium contains information about subsequent activity and that innovations in this premium generate broad macroeconomic contractions. Greenwood and Hanson (2013) relate the composition of bond issuance to investor sentiment and future corporate-bond returns. Most closely, López-Salido et al. (2017) find that periods in which credit risk is priced aggressively are followed by wider spreads and weaker activity. Evidence based on credit quantities reaches a similar conclusion: rapid credit growth predicts financial crises and deeper subsequent recessions (Schularick and Taylor, 2012; Jordà et al., 2013), while household-debt expansions forecast lower medium-run growth (Mian et al., 2017). These papers establish that easy credit conditions contain information about later contractions. We instead recover an orthogonal credit-spread shock and trace the entire response following the innovation. The initial expansion in output, investment, and hours is followed by a contraction, and stock prices display a similar reversal. The shock's explanatory power for the excess bond premium and its close relation to the high-yield issuance share connect this conditional response to the credit-sentiment evidence.

A second strand studies uncertainty as a source of recessions. The empirical analysis uses the broad macroeconomic uncertainty measure of Jurado et al. (2015). Bloom (2009) documents a sharp decline and rebound in activity after an uncertainty innovation, and Ludvigson et al. (2021) emphasizes the importance of separating exogenous uncertainty movements from the endogenous response of uncertainty to the business cycle. A central issue for this literature is comovement. Basu and Bundick (2017) show that higher uncertainty lowers output, consumption, investment, and hours in the data, whereas a standard flexible-price model predicts that precautionary labor supply raises hours, output, and investment. Nominal rigidity resolves this conflict by allowing lower demand to reduce production and factor demand. Leduc and Liu (2016) obtain a related contraction through nominal rigidity and labor-market search, while Arellano et al. (2019) use financial frictions to transmit greater firm-level volatility into lower input demand and tighter credit conditions. Basu et al. (2025) provide a real mechanism in which higher risk induces firms to substitute toward safer, less productive inputs. Relative to this work, we identify the uncertainty and credit-spread shocks jointly and show that they are not alternative labels for the same disturbance. The uncertainty shock produces the familiar recessionary comovement, but it does not generate the boom–bust dynamics associated with the orthogonal credit-spread shock.

On the model side, our paper is related to models in which financial conditions originate

or amplify aggregate fluctuations. In Jermann and Quadrini (2012), shocks to an enforcement constraint make firms' ability to borrow an independent source of business cycles. Financial constraints instead propagate shocks that originate elsewhere in Kiyotaki and Moore (1997) and Bernanke et al. (1999). Our model combines these roles. Uncertainty in the enforcement parameter directly tightens borrowing conditions and produces an immediate contraction. News about future productivity initially relaxes the same constraint; when the news is not realized, the debt accumulated during the anticipation phase becomes the source of the subsequent tightening. Unlike models that impose an always-binding collateral constraint, the constraint can be slack during the expansion and bind again when expectations reverse. This occasionally binding margin is central to the difference between the two shocks.

The paper is therefore also related to theories of endogenous boom–bust dynamics. Self-fulfilling fluctuations arise from financial constraints and strategic complementarities in Benhabib and Wang (2013), Liu and Wang (2014), and Azariadis et al. (2016). Beaudry et al. (2020) generate recurrent movements through a stochastic limit cycle around a locally unstable steady state. These approaches make cyclicality a feature of the propagation mechanism and therefore tend to pass it through to different disturbances. Our evidence instead requires conditional cyclicality: the credit-spread shock generates a boom and a bust, whereas the uncertainty shock generates a contraction followed by a return toward steady state. Brianti and Cormun (2024) also construct a model with shock-dependent dynamics, in which nonfundamental expectation shocks generate cycles while technology shocks do not. Bordalo et al. (2026) obtain credit reversals from diagnostic expectations: optimism in good times encourages debt and investment, and the later correction in beliefs raises spreads and depresses activity. Our mechanism shares the link between expectations, leverage, and reversal, but retains rational expectations. Optimism is represented by the anticipation of favorable news, and pessimism emerges when that news fails to materialize. The model must also account for the separate contractionary effect of macro uncertainty, which is absent from these conditional-cycle models.

Finally, the paper contributes to the computational literature on nonlinear dynamic models. Maliar et al. (2021) use neural networks to solve high- dimensional models by expressing equilibrium restrictions as nonlinear regression equations. Azinovic et al. (2022) develop Deep Equilibrium Nets, which learn policy functions by minimizing equilibrium residuals along simulated paths and can accommodate occasionally binding constraints. For estimation, Kase et al. (2025) integrate a global neural-network solution with the estimation of a nonlinear heterogeneous-agent model, while Wei and Jiang (2025) train a network to infer structural parameters from simulated moments. We use the neural network for a different purpose. Following the parameter-as-state strategy described by Scheidegger (2026), one network learns the equilibrium over both the economic state space and the relevant parameter region. We then estimate the model by matching model-implied and empirical impulse responses without resolving the model at every candidate parameter vector. This preserves the nonlinear transition between binding and slack

credit constraints while keeping the estimation criterion explicit.

2 Empirical Analysis

We begin the empirical analysis by documenting evidence of cyclical dynamics in U.S. macroeconomic data. In particular, hours worked has a pronounced spectral peak at business-cycle frequencies, and its medium-frequency autocorrelation changes sign as the horizon increases. To identify the shocks underlying these dynamics, we combine the max-share approach developed by Uhlig (2003) with the penalty-function approach of Caldara et al. (2016). This section first describes the data and then presents the two identification procedures. We next report the empirical findings in three stages. We recover the shock that accounts for the largest share of business-cycle variation in hours, examine its connection with macro uncertainty and credit spreads, and then separate an uncertainty shock from an orthogonal credit-spread shock. Finally, we use stock prices, the excess bond premium, and the high-yield issuance share to clarify the economic content of the latter shock.

2.1 Data

The baseline sample is quarterly and runs from 1973:I through 2024:IV. The VAR contains eight variables: output, consumption, investment, hours worked, the nominal interest rate, utilization-adjusted total factor productivity, the credit spread, and macro uncertainty. Output, consumption, investment, and hours are expressed as 100 times their natural logarithms. Output is real GDP per capita. Consumption consists of nondurable goods and services, while investment combines gross private domestic investment and consumer durables. Both aggregates are converted to real per-capita terms. Hours are total hours in the nonfarm business sector divided by the civilian noninstitutional population aged sixteen and older. These definitions follow the treatment of the national-accounts quantities in Angeletos et al. (2020).

The interest rate is the quarterly effective federal funds rate. Total factor productivity is the utilization-adjusted series constructed by Fernald (2014). The credit spread is the Gilchrist–Zakrajšek corporate bond spread, and macro uncertainty is the twelve-month-ahead measure of Jurado et al. (2015). The latter is a broad measure of uncertainty about future macroeconomic conditions rather than a financial-market volatility index. The credit spread and macro uncertainty are the two financial variables used in the baseline identification. Inflation is introduced only in the appendix specifications.

Three additional series help interpret the identified credit shock. The first is the real per-capita S&P index. The second is the excess bond premium of Gilchrist and Zakrajšek (2012), which removes the component of corporate spreads explained by expected defaults. The third is the share of high-yield issuance in total corporate bond issuance constructed by Greenwood and Hanson (2013). This high-yield ratio is a measure of the pricing of credit risk and is closely

related to the credit-market sentiment indicators studied by López-Salido et al. (2017). Because these series do not share a common sample, they enter as auxiliary variables rather than enlarging the baseline VAR.

For comparison with Angeletos et al. (2020), we also estimate a specification that augments the core macroeconomic variables with nonfarm business labor productivity and the labor share. This specification is used to verify that the hours-based business-cycle shock is not an artifact of omitting the productivity and income-distribution variables emphasized in their analysis. Appendix A gives the sources, transformations, and timing conventions for every series.

2.2 Empirical Method

Let x_t denote the $N \times 1$ vector of observables. The reduced-form VAR is

$$x_t = a + A_1 x_{t-1} + \cdots + A_p x_{t-p} + u_t, \qquad \mathbb{E}(u_t u_t') = \Sigma, \tag{1}$$

where the baseline lag length is $p = 2$. The VAR is estimated with a Minnesota prior and a Gibbs sampler, following Angeletos et al. (2020). We use 50,000 draws, discard the first 49,000, and calculate posterior medians and 68 percent posterior intervals from the retained draws.

Write $\Sigma = SS'$ and let q be a unit vector in the space of orthonormal structural innovations. The corresponding reduced-form impact vector is Sq. If B_h is the moving-average coefficient at horizon h, the impulse response is $B_h Sq$. Define

$$C(e^{-i\omega}) = \sum_{h=0}^{\infty} B_h S e^{-i\omega h}. \tag{2}$$

For variable k, the variation attributable to q over a frequency band $\mathcal{B} = [\underline{\omega}, \overline{\omega}]$ is proportional to

$$q' M_k(\mathcal{B}) q, \qquad M_k(\mathcal{B}) = \int_{\mathcal{B}} C_k(e^{-i\omega})' C_k(e^{-i\omega})\, d\omega, \tag{3}$$

where $C_k(\cdot)$ is row k of the transfer function. Dividing $M_k(\mathcal{B})$ by the total spectral variation of variable k over the same band expresses the objective as a variance share. The maximizing rotation is the leading eigenvector of the normalized matrix. When the target contains more than one variable, we maximize the equally weighted average of their normalized matrices. This normalization prevents a series with larger units or greater unconditional variance from determining the joint objective.

The main analysis sets $\mathcal{B} = [2\pi/40, 2\pi/6]$, corresponding to periodicities between 6 and 40 quarters. The lower-frequency boundary is wider than the conventional 6–32-quarter window used by Angeletos et al. (2020) because the peak in the spectral density of hours occurs near the upper end of that conventional window. Identification remains entirely in the frequency domain. The reported impulse responses are time-domain representations of the rotations selected by the

frequency-domain criterion; maximizing variation between 6 and 40 quarters is not equivalent to maximizing forecast-error variance over horizons 6 through 40.

2.2.1 The main business-cycle and joint shocks

The main business-cycle shock is the unit vector

$$q^H = \arg\max_{q'q=1} q' M_H(\mathcal{B}) q, \tag{4}$$

where H denotes hours worked. We orient the sign so that hours rise on impact. This is the hours-based counterpart of the maximum-share shocks studied by Angeletos et al. (2020). The distinction is that the target is hours and the VAR includes credit conditions and macro uncertainty.

To locate the information underlying q^H, we also identify

$$q^J = \arg\max_{q'q=1} q' \left[\tfrac{1}{2}\widetilde{M}_{GZ}(\mathcal{B}) + \tfrac{1}{2}\widetilde{M}_U(\mathcal{B})\right] q, \tag{5}$$

where tildes denote normalization by each variable's total variation over the band, GZ is the credit spread, and U is macro uncertainty. Comparing the shock series, impulse responses, and hours spectra associated with q^H and q^J establishes whether the main cyclical movement in hours is also the dominant common movement in uncertainty and credit spreads. This comparison is descriptive: neither rotation is assigned a structural label at this stage.

2.2.2 Separating uncertainty and credit conditions

The joint rotation cannot distinguish an uncertainty innovation from a credit innovation because the two observables move together. We therefore use a sequential identification, adapting the logic of Caldara et al. (2016) to the frequency-domain objective. The first rotation is

$$q^U = \arg\max_{q'q=1} q' \widetilde{M}_U(\mathcal{B}) q. \tag{6}$$

We call this the uncertainty shock. In the figures, its sign is chosen so that uncertainty falls on impact and real activity expands; the response to an increase in uncertainty is the negative of the reported response. The second rotation is restricted to the orthogonal complement of q^U:

$$q^C = \arg\max_{q'q=1,\ q'q^U=0} q' \left[\tfrac{1}{2}\widetilde{M}_{GZ}(\mathcal{B}) + \tfrac{1}{2}\widetilde{M}_U(\mathcal{B})\right] q. \tag{7}$$

The uncertainty direction has already been removed from the admissible set, so q^C isolates the remaining common credit-spread and uncertainty variation. We orient this shock so that output rises on impact. Its economic interpretation is based on its full set of responses and on the

auxiliary evidence below, not on the maximization criterion alone. In the appendix, we also report a rotation orthogonal to q^U that targets hours directly.

Orthogonality in (7) concerns the contemporaneous structural innovations. It does not impose a zero dynamic response of uncertainty to the second shock. The two shocks can therefore have different impact vectors and still generate responses in both the credit spread and uncertainty through the VAR's propagation mechanism.

2.2.3 Auxiliary variables

The S&P index, the excess bond premium, and the high-yield ratio are observed over different samples and are not included in the eight-variable VAR. For each auxiliary series z_t, we estimate

$$z_t = \alpha_z + \Gamma_{z,0} x_t + \Gamma_{z,1} x_{t-1} + \Gamma_{z,2} x_{t-2} + v_{z,t} \tag{8}$$

on the dates for which z_t is available. The response of z_t to the credit shock is then

$$\mathrm{IRF}_z(h) = \sum_{\ell=0}^{\min\{2,h\}} \Gamma_{z,\ell}\, \mathrm{IRF}_x(h-\ell; q^C). \tag{9}$$

This projection method follows the auxiliary-variable exercise in Basu et al. (2025). Posterior bands are computed by applying the projection to each retained VAR draw. Variance shares for an auxiliary variable include the residual variation in $v_{z,t}$ in the denominator, so the reported share is not mechanically inflated by the fitted component of (8).

2.3 Hours in the Frequency and Time Domains

Figure 1 reports the two facts that motivate the identification exercise. The spectral density of hours has a sharp maximum at a periodicity of approximately 38 quarters. The peak is present in the level series and across the high-pass-filtered series shown by the gray envelope. The right panel gives the corresponding time-domain evidence. After removing fluctuations with periodicities longer than 80 quarters, the autocorrelation of hours is positive at short lags, crosses zero after roughly eight quarters, remains negative through approximately lag 29, and reaches its trough around lags 15–17. Thus, the concentration of variation at business-cycle frequencies is accompanied by a change in the sign of serial dependence.

The connection between these two observations follows from the spectral representation. For a covariance-stationary process, the spectral density is the Fourier transform of its autocovariances:

$$f_H(\omega) = \frac{1}{2\pi}\left[\gamma_0 + 2\sum_{m=1}^{\infty} \gamma_m \cos(m\omega)\right], \qquad \omega \in [0,\pi]. \tag{10}$$

If every γ_m for $m \geq 1$ were nonnegative, then $f_H(\omega) \leq f_H(0)$ because $\cos(m\omega) \leq 1$. A spectral

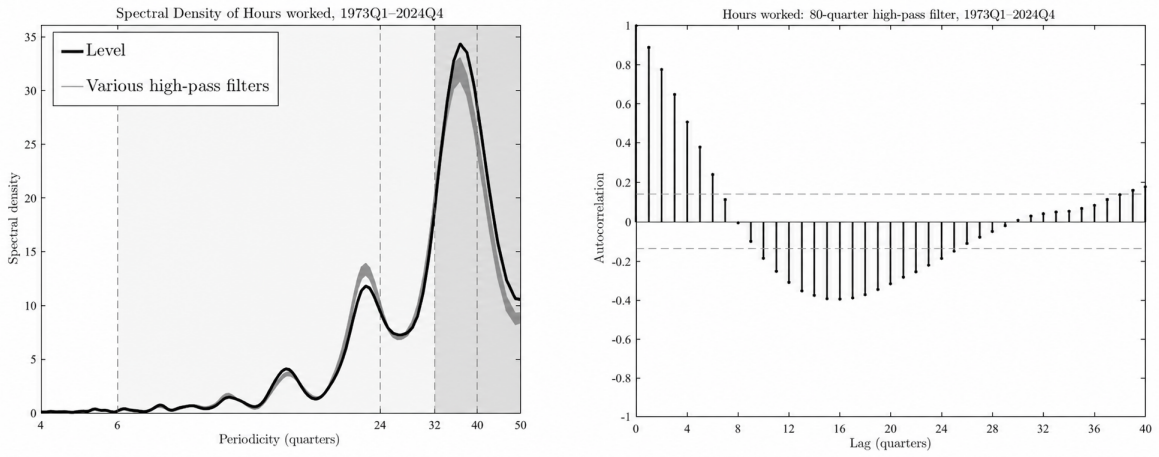


Figure 1: Hours Worked in the Frequency and Time Domains

Notes. The left panel reports the spectral density of log hours per capita from 1973:I to 2024:IV. The black line is the spectrum of the level series; the gray region spans spectra obtained under the reported high-pass filters. Dashed vertical lines mark periodicities of 6, 24, 32, and 40 quarters. The right panel reports sample autocorrelations after applying an 80-quarter high-pass filter. Dashed horizontal lines are approximate 95 percent white-noise bounds.

maximum away from frequency zero therefore requires at least one negative autocovariance. The sign change in the right panel of Figure 1 provides the corresponding time-domain evidence directly.

The implication for propagation is especially transparent in the context of a single shock. Suppose hours admit the invertible moving-average representation

$$H_t = \sum_{j=0}^{\infty} \phi_j \varepsilon_{t-j}, \qquad \mathbb{E}(\varepsilon_t^2) = \sigma_\varepsilon^2, \tag{11}$$

where ϕ_j is the response of hours at horizon j. Its autocovariance at lag m is

$$\gamma_m = \sigma_\varepsilon^2 \sum_{j=0}^{\infty} \phi_j \phi_{j+m}. \tag{12}$$

If the impulse-response coefficients were all weakly positive, or all weakly negative, every product in (12) would be nonnegative. Consequently, a negative autocovariance requires coefficients of opposite signs at some horizons: the impulse response must cross the horizontal axis. This is only a necessary condition. It neither determines the horizon of the reversal nor identifies the underlying shock. The empirical analysis below asks why this sign-changing propagation is difficult to see in the response to the shock that accounts for most of the business-cycle variation in hours.

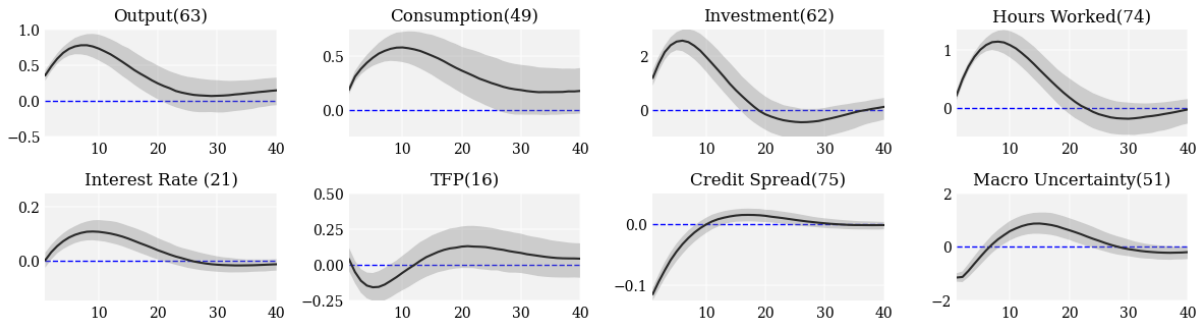


Figure 2: Impulse Responses to the Main Business-Cycle Shock

Notes. Solid lines are posterior median responses and shaded regions are posterior credible bands. The shock maximizes the variance share of hours over periodicities from 6 to 40 quarters and is normalized to raise hours on impact. Numbers in parentheses report the percentage of each variable's variation over that band attributed to the shock. The horizontal axis is in quarters.

2.4 Empirical Results

2.4.1 The main business-cycle shock

Figure 2 reports the responses to the shock that maximizes the 6–40-quarter variance share of hours. The shock produces an expansion in all four activity variables. Output, consumption, investment, and hours rise together, peak within the first ten quarters, and then return gradually toward zero. The shock accounts for 74 percent of the variation in hours over the target band, as reported in parentheses in the panel title. Its shares of output, consumption, and investment are 63, 49, and 62 percent, respectively.

The median responses of investment and hours eventually fall slightly below zero, but their negative portions are small relative to the initial expansion and are imprecisely estimated. The response therefore does not display a clear boom followed by a statistically distinct bust. The financial variables are more revealing. The credit spread and macro uncertainty fall sharply on impact and subsequently overshoot, and the shock explains 75 percent of the business-cycle variation in the spread and 51 percent of that in uncertainty. The main business-cycle shock is therefore closely associated with joint movements in these two variables.

Figure 3 shows the frequency-domain counterpart. The first principal component, which is the main business-cycle shock, accounts for most of the hours spectrum at long business-cycle periodicities. The second orthogonal component contributes comparatively little. Their sum retains a pronounced peak, although its maximum lies somewhat to the right of the peak in the data. Hence the hours-maximizing rotation reproduces the concentration of variance at business-cycle frequencies even though its median impulse response contains only a weak and delayed negative phase. This is the empirical puzzle: the cyclical component is clear in the spectrum but muted in the impulse response of the dominant rotation.

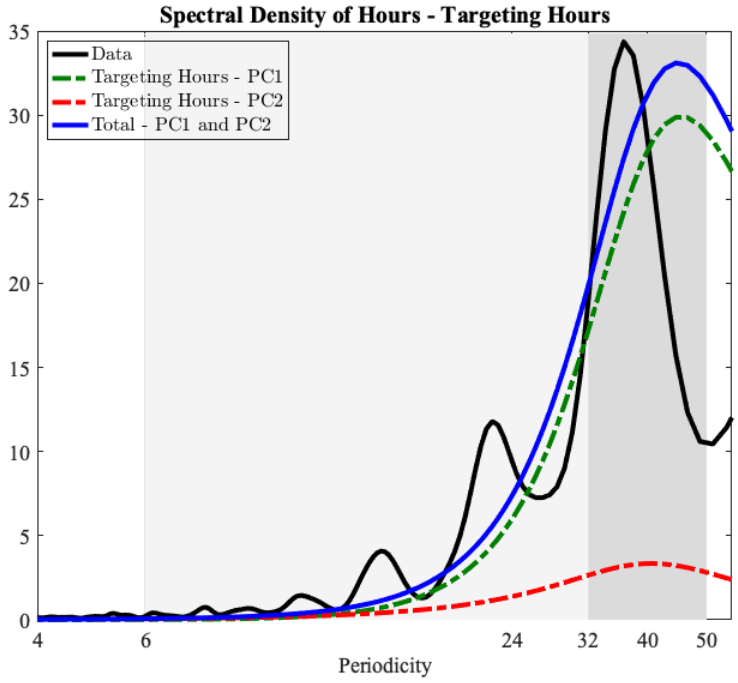


Figure 3: Hours Spectrum under the Hours-Maximizing Rotation

Notes. The black line is the spectral density of hours in the data. The green dashed line is the spectrum attributable to the first principal component of the hours objective, and the red dashed line is the spectrum attributable to the second component. The blue line is their sum. The shaded region marks periodicities from 32 to 50 quarters.

2.4.2 Uncertainty and credit conditions within the MBC shock

Figures 4 and 5 repeat the exercise for the rotation that jointly maximizes the normalized business-cycle variation of the credit spread and macro uncertainty. Its responses are nearly indistinguishable from those of the hours-maximizing shock. The four activity variables rise together, while the credit spread and uncertainty fall on impact and later overshoot. The associated hours spectrum has the same broad shape and is again dominated by the first component. More directly, the two estimated shock series have a correlation of 0.97. The dominant movement in hours is therefore also the dominant common movement in uncertainty and credit spreads.

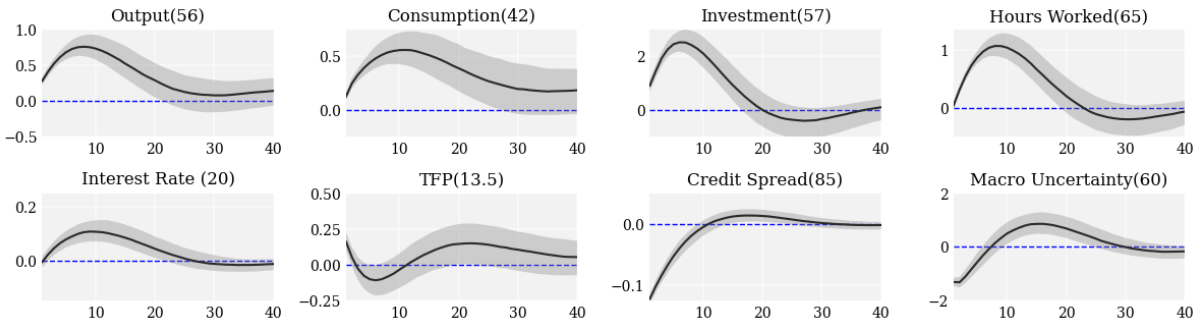


Figure 4: Impulse Responses to the Joint Uncertainty–Credit-Spread Shock

Notes. Solid lines are posterior median responses and shaded regions are posterior credible bands. The shock jointly maximizes the normalized variance shares of the credit spread and macro uncertainty over periodicities from 6 to 40 quarters. Numbers in parentheses report variance shares over that band. The horizontal axis is in quarters.

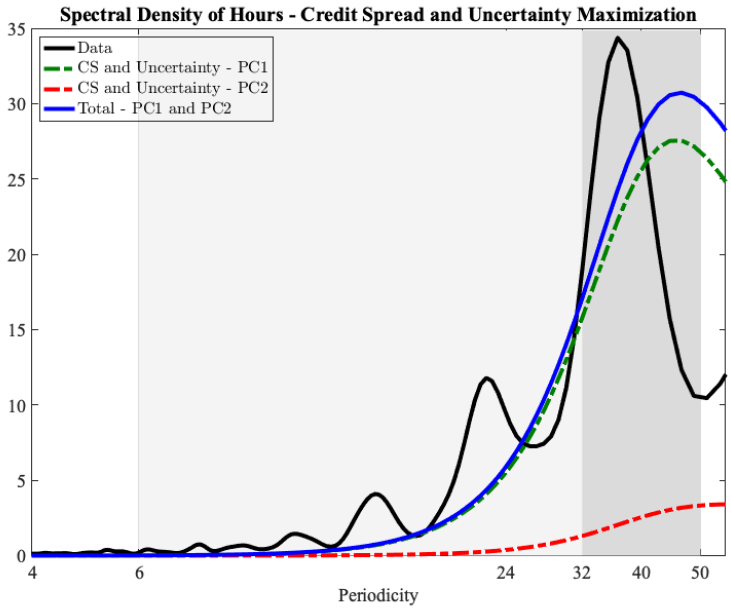


Figure 5: Hours Spectrum under the Joint Rotation

Notes. The black line is the spectral density of hours in the data. The green and red dashed lines are the spectra attributable to the first and second principal components of the joint credit-spread and uncertainty objective. The blue line is their sum. The shaded region marks periodicities from 32 to 50 quarters.

The similarity does not imply that a single structural disturbance drives both financial variables. The joint criterion selects their dominant common direction, which can combine innovations with different propagation. We use the sequential restrictions in equations (6) and (7) to separate that direction.

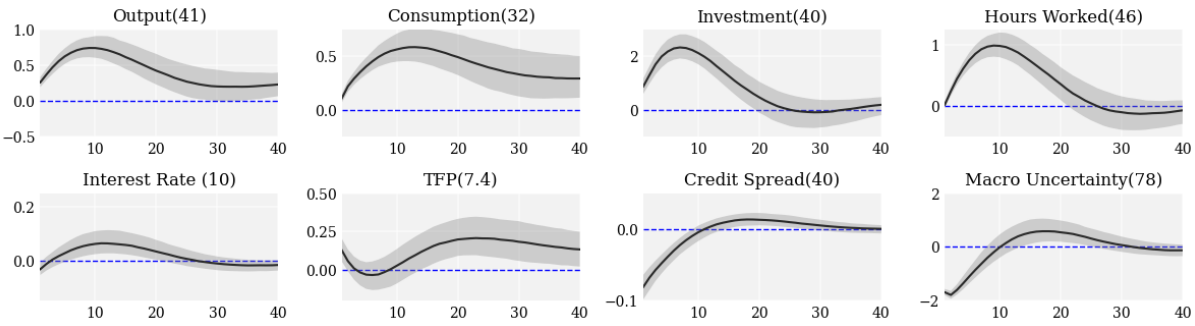


Figure 6: Impulse Responses to the Uncertainty Shock

Notes. Solid lines are posterior median responses and shaded regions are posterior credible bands. The shock maximizes the variance share of macro uncertainty over periodicities from 6 to 40 quarters. Its sign is normalized so that uncertainty falls on impact. Numbers in parentheses report variance shares over the target band. The horizontal axis is in quarters.

Figure 6 reports the first rotation, with the sign normalized as a decline in uncertainty. Output, consumption, investment, and hours all expand. Their responses decay gradually and exhibit no economically or statistically pronounced contraction. Reversing the sign gives the usual uncertainty experiment: an increase in macro uncertainty contracts all four activity variables and the effects dissipate without a subsequent boom. This shock accounts for 78 percent of the target-band variation in uncertainty but only 40 percent of the credit-spread variation.

The second rotation, shown in Figure 7, is orthogonal to the uncertainty shock and maximizes the remaining joint variation in the credit spread and uncertainty. Its propagation is different. A sharp decline in the credit spread is followed by a broad expansion: output, consumption, investment, and hours rise together for roughly the first 12–15 quarters. Each response then changes sign. The contraction is most pronounced for output, investment, and hours, while the decline in consumption is weaker and less precisely estimated. The trough occurs after approximately 20–25 quarters, followed by a gradual recovery.

This shock explains 53 percent of the business-cycle variation in the credit spread but only 12 percent of the variation in uncertainty. Uncertainty can nevertheless respond dynamically because the orthogonality restriction applies to the contemporaneous innovations, not to subsequent VAR propagation. The second rotation thus isolates a credit-related boom–bust component that is largely obscured when uncertainty and credit conditions are combined in a single dominant shock.

Figure 8 clarifies how the two rotations contribute to the hours spectrum. The uncertainty component places relatively more mass at periodicities beyond the center of the business-cycle window. The orthogonal credit component rises more sharply around the observed peak and reaches its maximum near 40 quarters. Separating the two components therefore uncovers the sign-changing propagation that is difficult to see in the MBC and joint-shock impulse responses.

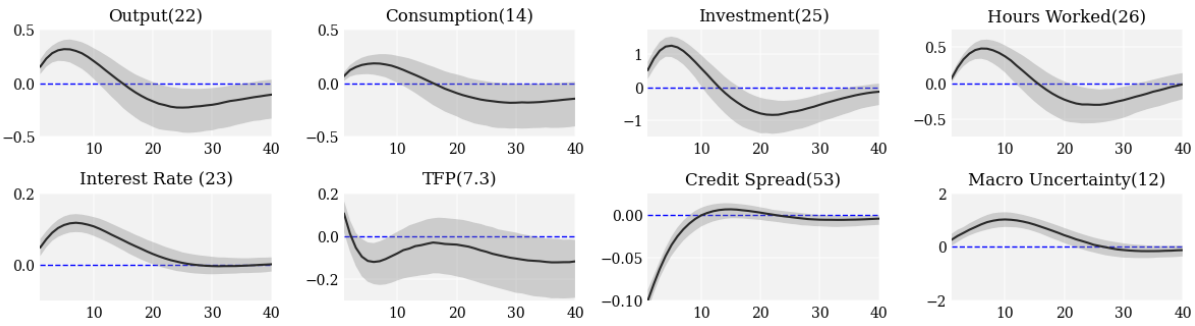


Figure 7: Impulse Responses to the Shock Orthogonal to Uncertainty

Notes. Solid lines are posterior median responses and shaded regions are posterior credible bands. The shock is orthogonal to the uncertainty shock and maximizes the remaining joint variation in the credit spread and uncertainty over periodicities from 6 to 40 quarters. It is normalized to lower the credit spread on impact. Numbers in parentheses report variance shares over the target band. The horizontal axis is in quarters.

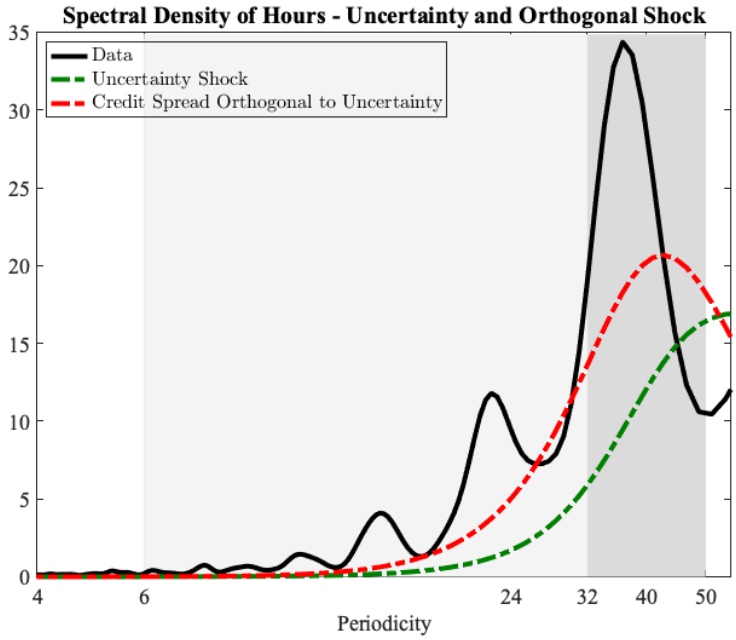


Figure 8: Hours Spectra of the Uncertainty and Orthogonal Credit Shocks

Notes. The black line is the spectral density of hours in the data. The green dashed line is the spectrum attributable to the uncertainty shock. The red dashed line is the spectrum attributable to the shock orthogonal to uncertainty that targets the remaining joint credit-spread and uncertainty variation. The shaded region marks periodicities from 32 to 50 quarters.

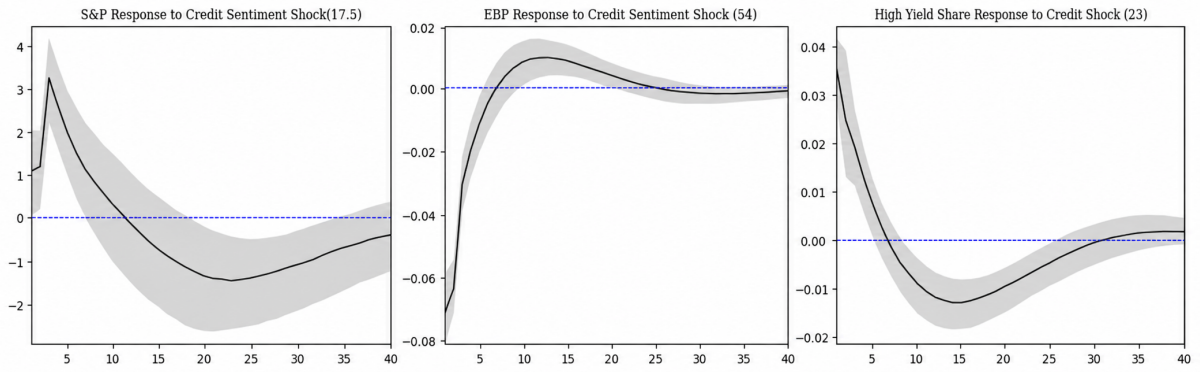


Figure 9: Financial Responses to the Credit Sentiment Shock

Notes. Panels report the projected responses of the real per-capita S&P index, the excess bond premium, and the high-yield issuance share to the shock in Figure 7. Solid lines are posterior medians and shaded regions are posterior credible bands. Numbers in parentheses report the variance shares attributed to the shock. The horizontal axis is in quarters.

2.4.3 Interpreting the orthogonal shock

Figure 9 uses the auxiliary projections to examine the financial content of the second rotation. The S&P index rises sharply and then falls below its initial level, producing the same boom–bust pattern as real activity. The excess bond premium drops on impact, subsequently becomes positive, and then returns toward zero. The shock explains 54 percent of the variation in the excess bond premium captured by the auxiliary projection. The high-yield issuance share also rises initially, turns negative after the expansion, and recovers only gradually.

These responses connect the shock to the credit-market sentiment evidence of Greenwood and Hanson (2013) and López-Salido et al. (2017). The combination of compressed risk compensation, elevated risky issuance, rising equity values, and a later reversal is consistent with an improvement in credit sentiment followed by retrenchment. We therefore refer to q^C as the *credit sentiment shock*. This label is an interpretation of its joint macroeconomic and financial responses; it is not imposed by the identifying restriction.

2.5 Discussion: Implications for the Model

The empirical results impose two distinct requirements on the propagation mechanism. The credit sentiment shock generates a sign-changing response: an expansion in output, consumption, investment, and hours is followed by a contraction. An increase in macro uncertainty instead produces a broad contraction that gradually returns to steady state without an economically meaningful sign reversal. Thus, the model must generate comovement among the four quantities after both shocks while allowing the shape of the response to depend on the disturbance that initiates it. Cyclical propagation cannot be an unconditional property of every shock that enters

the economy.

One class of explanations generates cycles through indeterminacy and self-fulfilling beliefs. In the early sunspot literature, increasing returns or other strategic complementarities make equilibrium activity respond to innovations in beliefs (Benhabib and Farmer, 1994). Financial constraints provide related mechanisms without requiring increasing returns at the firm level. Expected firm value, borrowing capacity, markups, and the allocation of capital can reinforce one another, allowing belief shocks to produce persistent and potentially oscillatory movements in aggregate activity (Benhabib and Wang, 2013; Liu and Wang, 2014; Azariadis et al., 2016). These mechanisms can produce the boom–bust response found for the credit sentiment shock. Their cyclical dynamics, however, are typically embedded in the equilibrium law of motion. Once the economy is in the indeterminate region, sunspot and fundamental innovations are propagated through the same endogenous state dynamics unless additional restrictions are placed on equilibrium selection or on the loading of each shock. Cyclicality is therefore not intrinsically linked to the economic nature of the disturbance, whereas the empirical results require precisely this conditionality.

An important exception is Brianti and Cormun (2024). Their model allows expectation shocks to generate boom–bust dynamics while technology shocks produce non-oscillatory responses. The distinction arises because the response of beliefs to the technology innovation can be strong enough for the fundamental component to dominate the endogenous expectation dynamics. This establishes that a self-fulfilling equilibrium can, with a suitable shock-specific loading, generate conditional cyclicality. The alternative propagation in that model is nevertheless tied to a technology disturbance. Here, the non-oscillatory contraction is generated by macro uncertainty, and it features comovement in all four quantities. A flexible-price RBC mechanism does not deliver that response to uncertainty: precautionary saving lowers consumption but raises labor supply, output, and investment (Basu and Bundick, 2017). The relevant empirical restriction is therefore not merely to distinguish an expectation shock from TFP, but to reconcile a credit-driven cycle with a contractionary uncertainty response.

A second class of explanations places the economy near a stochastic limit cycle. In Beaudry et al. (2020), financial complementarities and a locally unstable steady state generate recurrent boom–bust dynamics even when the exogenous disturbances themselves are not persistent. This approach preserves rational expectations and provides a direct account of a spectral peak. At the same time, recurrence is a property of the deterministic core of the economy. Shocks perturb the economy around the same underlying orbit, so their subsequent responses naturally inherit its cyclical propagation. This makes it difficult to obtain a pronounced cycle after the credit sentiment shock while shutting down the same reversal after the uncertainty shock.

Diagnostic-expectations models provide a third route. In Bordalo et al. (2026), agents overreact to good news, credit and investment expand too far, and the later fading of optimism produces a financial reversal. Because the strength of the reversal depends on news and the state

of the economy, this framework can accommodate both cyclical and non-oscillatory responses. Its application to uncertainty faces a separate comovement problem. In a neoclassical environment, an increase in uncertainty already strengthens the precautionary-saving motive. If diagnostic beliefs magnify the perceived adverse states, this force becomes stronger: consumption falls while desired saving and labor supply rise. Without an additional mechanism that shifts labor demand and aggregate expenditure, overreaction makes a simultaneous contraction in consumption, investment, output, and hours harder to obtain.

These comparisons narrow the relevant modeling task. The reversal should arise from the interaction of optimistic credit conditions with subsequent disappointment, rather than from an oscillatory law of motion shared by every shock. Macro uncertainty should operate through the same financial environment but generate a contractionary, mean-reverting response. The next section develops a rational-expectations model in which anticipated productivity news relaxes the financing constraint, its non-realization reverses that expansion, and uncertainty about enforcement tightens financing conditions without first generating a credit boom. Nominal wage rigidity then permits both disturbances to produce the four-variable comovement observed in the data.

3 The Model

3.1 Overview

The model is designed to account for two empirical responses documented in the previous section: a drop in credit spread (or excess bond premia) led to expansion that later reversed to contraction of the economy, and a contraction following an increase in macro uncertainty. In both cases, output, consumption, investment, and hours move in the same direction. This co-movement is a demanding restriction for one-sector models (Basu and Bundick, 2017; Angeletos et al., 2020). Since both innovations generate similar responses in the credit spread, the analysis asks whether a common financing mechanism can generate both responses.

The economy contains infinitely lived households, competitive firms, and a central bank. Households consume, supply differentiated labor, and hold one-period nominal bonds. Firms own capital, hire labor, make investment, issue debt, and make equity payouts. The central bank sets the short nominal rate with a Taylor rule. Goods prices are flexible; nominal wages adjust slowly.

Households deal with wage rigidity, while firms navigate financial constraints. The financial setup rests on three key elements. First, firms need to borrow during each period to cover labor wages before they receive any revenue, referred to as working capital loans. However, a constraint limits how much firms can borrow for this purpose, tying it to the value of capital that can be pledged, after accounting for outstanding debt. Second, if firms adjust their equity payouts away from their long-term target, they face convex adjustment costs. Lastly, the model incorporates tax benefits for debt and costs of adjusting payouts, as discussed by Jermann and Quadrini

(2012), along with a changing enforcement limit on wage bills and its uncertainty, following Chatterjee et al. (2025). These costs keep firms from easily using equity to make up for drops in borrowing capacity. As a result, credit conditions end up influencing hiring decisions through the value firms place on their internal funds.

One further feature links investment to borrowing capacity. Convex investment adjustment costs make the value of installed capital respond to the current investment rate. Because the borrowing limit depends on this value, investment can affect borrowing capacity immediately even though the capital stock is predetermined. Together with the nominal wage rigidity described above, this link allows changes in credit conditions to move output, consumption, investment, and hours in the same direction.

The model is perturbed by productivity surprises, news about future productivity, and innovations to enforcement and its volatility. News arrives with an observed magnitude and horizon, but is either confirmed or denied at maturity. Anticipation and disappointment provide a rational-expectations representation of credit-market optimism and pessimism. The enforcement constraint may become slack during anticipation and bind again after denial; its status is determined in equilibrium. The uncertainty experiment raises the dispersion of enforcement innovations, following Chatterjee et al. (2025). We solve and estimate the model using a deep equilibrium network (DEQN) (Azinovic et al., 2022; Scheidegger, 2026).

3.2 Households

Preferences and budget constraint. There is a continuum of households indexed by $j \in [0,1]$, each supplying a differentiated variety of labor. Household j chooses consumption, bond holdings and its own nominal wage to maximise

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{\left(c_t(j) - \iota c_{t-1}(j)\right)^{1-\sigma} - 1}{1-\sigma} - \alpha \frac{n_t(j)^{1+\nu}}{1+\nu} \right], \tag{13}$$

where $\beta \in (0,1)$ is the discount factor, $\sigma > 0$ governs the curvature of the consumption aggregate, $\nu > 0$ is the inverse Frisch elasticity, and $\iota \in [0,1)$ is the degree of habit persistence. Habit is internal: the household recognizes that raising consumption today raises the reference stock against which its own utility is evaluated tomorrow, and internalizes that effect. This differs from the external specification in the estimated model of Jermann and Quadrini (2012) and adds a forward-looking term to the marginal utility of wealth, as recorded below.

Consumption, investment, dividends and taxes are real; bonds and wages are nominal and are deflated by the current price level. The budget constraint is

$$c_t(j) + \frac{B_{t+1}(j)}{P_t(1+i_t)} = \left(1 - \Theta_{w,t}(j)\right) \frac{W_t(j)}{P_t} n_t(j) + \frac{B_t(j)}{P_t} + d_t - T_t, \tag{14}$$

where $B_t(j)$ is the nominal face value contracted previously and due at t, i_t is the net nominal

interest rate on a one-period bond, d_t is the real dividend received from firm ownership, T_t is a real lump-sum tax, and $\Theta_{w,t}(j)$ is the wage adjustment cost defined below. There is no inflation-indexed bond. Because all debt is nominal, unanticipated inflation revalues outstanding liabilities in real terms, so the model carries a debt-deflation channel absent from purely real formulations of the same financing block.

labor varieties and wage setting. Household j sells its variety to a competitive labor packer that assembles the aggregate input used by firms. The packer's cost minimization delivers the downward-sloping demand curve

$$n_t(j) = \left(\frac{W_t(j)}{W_t}\right)^{-\eta_w} n_t, \tag{15}$$

where W_t is the aggregate nominal wage index, n_t is aggregate labor demand, and $\eta_w > 1$ is the elasticity of substitution across varieties, which implies a desired wage markup $\mu_w \equiv \eta_w/(\eta_w - 1)$.

Households are monopolistic suppliers of their own variety and face a real resource cost, borne as a share of their own wage bill, when they change their posted nominal wage:

$$\Theta_{w,t}(j) = \frac{\phi_w}{2}\big(\Pi_{w,t}(j) - 1\big)^2, \qquad \Pi_{w,t}(j) \equiv \frac{W_t(j)}{W_{t-1}(j)}, \tag{16}$$

with $\phi_w \geq 0$ governing the degree of stickiness. Two features of (16) matter. The cost applies to *nominal* wage growth; a cost in real wage growth would not create a nominal friction. Symmetry and convexity imply a common wage, so wage dispersion is not an additional state. The specification yields a forward-looking wage Phillips curve. Born and Pfeifer (2020) provide the mapping from ϕ_w to the average contract duration in an equivalent staggered-contract economy, which we use to assess the calibration in Section 3.8.

Optimality conditions. Let $\Lambda_t(j)$ denote the multiplier on household j's budget constraint and define its flow marginal felicity as $X_t(j) \equiv [c_t(j) - \iota c_{t-1}(j)]^{-\sigma}$. In a symmetric equilibrium, consumption, bond holdings, and their associated multipliers are common across households. We therefore drop the j index from these objects from this point on. Because habit is internal, the marginal utility of wealth nets off the future cost of a higher habit stock,

$$\Lambda_t = X_t - \iota\beta\,\mathbb{E}_t[X_{t+1}]\,, \tag{17}$$

so that $\Lambda_t < X_t$ whenever $\iota > 0$: the household perceives itself as gaining less from an extra unit of consumption than the flow term alone would suggest. The stochastic discount factor over real payoffs is $m_{t,t+1} \equiv \beta\Lambda_{t+1}/\Lambda_t$, and the pricing condition for the nominal bond is

$$1 = (1 + i_t)\,\mathbb{E}_t\left[\frac{m_{t,t+1}}{\Pi_{t+1}}\right], \qquad \Pi_{t+1} \equiv \frac{P_{t+1}}{P_t}. \tag{18}$$

Equation (18) holds exactly under uncertainty and is the only intertemporal asset-pricing condition in the model.

Imposing symmetry across households and defining the marginal rate of substitution $\mathrm{mrs}_t \equiv \alpha n_t^{\nu}/\Lambda_t$ and the real wage $w_t \equiv W_t/P_t$, the optimal wage satisfies

$$\phi_w\big(\Pi_{w,t}-1\big)\Pi_{w,t} = (1-\eta_w)\big(1-\Theta_{w,t}\big) + \eta_w \frac{\mathrm{mrs}_t}{w_t} + \mathbb{E}_t\left[m_{t,t+1}\,\phi_w\big(\Pi_{w,t+1}-1\big)\Pi_{w,t+1}^2\,\frac{n_{t+1}}{n_t}\,\frac{1}{\Pi_{t+1}}\right]. \tag{19}$$

Equation (19) is the nonlinear wage Phillips curve. When $\phi_w = 0$ it collapses to the static markup rule $w_t = \mu_w\,\mathrm{mrs}_t$ of Erceg et al. (2000), which we use as a numerical check on the solution. When $\phi_w > 0$, the real wage need not track the marginal rate of substitution period by period, allowing labor demand to determine hours over the horizons of interest.

3.3 Firms

The firm sector contains the model's financial friction.

Technology and capital accumulation. Firms are perfectly competitive and produce with a constant-returns Cobb–Douglas technology,

$$y_t = z_t k_t^{\theta} n_t^{1-\theta}, \tag{20}$$

where z_t is aggregate productivity, k_t is the predetermined capital stock and θ is the capital share. There is no price rigidity and no markup on the goods side; every nominal rigidity in the model resides in wages. Capital accumulates subject to a convex cost in the level of the investment rate,

$$k_{t+1} = (1-\delta)k_t + I_t - \frac{\phi}{2}\left(\frac{I_t}{k_t} - \delta\right)^2 k_t, \tag{21}$$

following Chatterjee et al. (2025). The cost is specified in terms of I_t/k_t rather than in its growth rate, which keeps the firm's individual state two-dimensional and, as Section 3.7 shows, delivers a transparent closed-form link between the investment rate and the value of collateral.

Financing. Firms use two instruments. Intertemporal debt is nominal and has face value B_{t+1}, and is preferred to equity because interest is tax-deductible: given the policy rate i_t, the effective gross rate paid by the firm is

$$R_t = 1 + i_t(1-\tau), \tag{22}$$

where τ is the deductibility parameter. The resulting subsidy is financed by the lump-sum tax $T_t = B_{t+1}/(P_t R_t) - B_{t+1}/\big(P_t(1+i_t)\big)$ levied on households, so the tax shield is a pure financing wedge and not a transfer of resources into or out of the economy. In addition, firms take an

intraperiod working-capital loan to cover the wage bill, which is paid before revenues arrive and repaid within the period at no interest.

The equity payout is costly to adjust. Given a payout d_t, the resource cost borne by the firm is $d_t + \kappa(d_t - \bar{d})^2$, where $\bar{d}$ is the long-run payout target and $\kappa \geq 0$. Following Jermann and Quadrini (2012), this should not be read as a literal pecuniary cost but as a parsimonious way of capturing the speed with which firms can rebalance between debt and equity when financial conditions change; it is consistent both with the dividend-smoothing behaviour documented since Lintner (1956) and with increasing marginal costs of equity issuance. The parameter κ turns out to be the pivotal one: it is what prevents a change in pledgeable value from being absorbed silently by the financial structure. Collecting these elements, the firm's flow of funds in real terms is

$$y_t + \frac{B_{t+1}}{P_t R_t} = d_t + \kappa \left(d_t - \bar{d}\right)^2 + w_t n_t + I_t + \frac{B_t}{P_t}. \tag{23}$$

The enforcement constraint. Debt contracts are imperfectly enforceable. Lenders extend the working-capital loan only up to the value they expect to recover, which is a fraction ζ_t of the collateral remaining after the first claim of intertemporal debt-holders has been met:

$$\zeta_t \left[Q_t k_t - \frac{B_{t+1}}{P_t(1 + i_t)} \right] \;\geq\; w_t n_t, \tag{24}$$

with associated multiplier $\mu_t \geq 0$ and complementary slackness. Here Q_t is the price at which a unit of installed capital is valued in the contract, measured in goods, and k_t is the stock available when the contract is signed. Three features of (24) deserve comment. First, the constraint is written on the wage bill, not on output, so the credit wedge attaches directly to the hiring margin. Second, it is the predetermined stock k_t that serves as collateral, not the choice k_{t+1}: the firm cannot relax today's borrowing limit by investing more today, and the value of extra capital in loosening future constraints therefore enters the capital Euler equation through the envelope condition rather than as a contemporaneous subsidy to investment. Third, the current market value of newly issued intertemporal debt is deducted, so leverage and pledge-ability compete for the same collateral. The parameter ζ_t is common across firms and stochastic; its innovations are the credit shocks of the model, and we interpret movements in ζ_t as changes in the recover-ability of collateral rather than as shifts in investor beliefs.

Writing the unused borrowing capacity as

$$S_t \equiv \zeta_t \left[Q_t k_t - \frac{B_{t+1}}{P_t(1 + i_t)} \right] - w_t n_t, \qquad S_t \geq 0, \quad \mu_t \geq 0, \quad \mu_t S_t = 0, \tag{25}$$

makes the two regimes explicit. When $S_t > 0$, the constraint is slack and $\mu_t = 0$; when $\mu_t > 0$, it binds and $S_t = 0$. Binding in the deterministic steady state does not impose binding along stochastic paths.

Collateral valuation. It remains to say how lenders price the pledged stock. Let q_t be the multiplier on (21), which measures the marginal value of installed capital, and λ_t the multiplier on (23), which measures the marginal value of one good. Their ratio is the marginal value of capital expressed in goods, and we assume that lenders price collateral at this value:

$$Q_t = \frac{q_t}{\lambda_t}. \tag{26}$$

Equation (26) is a valuation rule imposed on the symmetric equilibrium, not an identity derived from an explicit liquidation market, and it does not make Q_t exogenous: the price is determined jointly with investment, payouts and the remaining equilibrium objects. An individual firm takes Q_t as given, and (26) is applied only after the firm's optimality conditions have been derived, so that no firm treats the aggregate valuation rule as a direct dependence of the contract price on its own choices. When payouts adjust freely, $\kappa = 0$ implies $\lambda_t \equiv 1$ and hence $Q_t = q_t$.

The firm's problem. The individual state is (k_t, B_t), with aggregate states suppressed in the time subscript of the value function. The firm chooses $\{d_t, n_t, I_t, k_{t+1}, B_{t+1}\}$ to maximise cum-dividend value

$$V_t(k_t, B_t) = d_t + \mathbb{E}_t[m_{t,t+1} V_{t+1}(k_{t+1}, B_{t+1})] \tag{27}$$

subject to (20), (21), (23) and (24), discounting with the owners' stochastic discount factor and taking w_t, i_t, P_t and Q_t as given.

Optimality conditions. The payout condition is $\lambda_t = \left[1 + 2\kappa(d_t - \bar{d})\right]^{-1}$, so λ_t falls below unity exactly when the firm pays out more than its target. The labor choice yields the central static condition of the model,

$$w_t \left(1 + \frac{\mu_t}{\lambda_t}\right) = (1 - \theta)\frac{y_t}{n_t}. \tag{28}$$

labor is hired to the point where its marginal product equals the wage augmented by a financing wedge, μ_t/λ_t, which is the shadow cost of the collateral constraint expressed per unit of internal funds. A higher wedge shifts labor demand inward at a given wage and a lower wedge does the reverse. Equation (28) is the only condition in which the shadow cost of the constraint bears directly on a contemporaneous real allocation, although, as Section 3.7 shows, what moves the wedge is determined elsewhere in the system. It is the model's counterpart to the labor wedge in Jermann and Quadrini (2012) and Chatterjee et al. (2025), generalized here to account for the marginal value of internal funds.

The investment condition is $\lambda_t = q_t\left[1-\phi(I_t/k_t-\delta)\right]$. Capital and debt satisfy

$$q_t = \mathbb{E}_t\left[m_{t,t+1}\left(q_{t+1}\Xi_{t+1} + \lambda_{t+1}\theta\frac{y_{t+1}}{k_{t+1}} + \underbrace{\mu_{t+1}\zeta_{t+1}Q_{t+1}}_{\text{collateral payoff}}\right)\right], \tag{29}$$

$$\mathbb{E}_t\left[\frac{m_{t,t+1}\lambda_{t+1}}{\Pi_{t+1}}\right] = \frac{\lambda_t}{R_t} - \frac{\mu_t\zeta_t}{1+i_t}, \tag{30}$$

where $\Xi_{t+1} \equiv (1-\delta) + \phi\left(\frac{I_{t+1}}{k_{t+1}}-\delta\right)\frac{I_{t+1}}{k_{t+1}} - \frac{\phi}{2}\left(\frac{I_{t+1}}{k_{t+1}}-\delta\right)^2$ collects the standard user-cost terms and we have used $\varphi_t = \lambda_t$, the optimality condition for output.

Two terms in (29)–(30) are specific to this environment. In (29) the marginal unit of capital pays not only its marginal product and its resale value but also a collateral dividend $\mu_{t+1}\zeta_{t+1}Q_{t+1}$: capital carried into the next period has an additional payoff in states where it relaxes a binding financing constraint. This payoff is zero in slack states. In (30) the marginal cost of debt is raised by $\mu_t\zeta_t/(1+i_t)$, because a unit of new borrowing consumes pledgeable collateral one for one. The tax shield pushes the firm towards more leverage and the enforcement constraint pushes back; Section 3.7 shows that this tension makes the constraint bind at the deterministic steady state.

3.4 The Nominal Block and Monetary Policy

Goods prices are flexible, so there is no price Phillips curve. Price inflation is instead recovered from the definition of the real wage: since $w_t \equiv W_t/P_t$,

$$\Pi_t = \Pi_{w,t}\cdot\frac{w_{t-1}}{w_t}. \tag{31}$$

Equation (31) is an accounting identity and holds regardless of any stickiness assumption. It links the real wage determined by (28) and nominal wage inflation determined by (19) to the inflation rate in the policy rule. Price inflation falls when the real wage rises faster than the nominal wage.

The central bank follows a rule in current inflation and the level of output,

$$1+i_t = (1+i_{ss})\left(\frac{\Pi_t}{\Pi^*}\right)^{\phi_\pi}\left(\frac{y_t}{y_{ss}}\right)^{\phi_y}, \tag{32}$$

with target $\Pi^* = 1$ and $1+i_{ss} = 1/\beta$. Output is measured relative to its deterministic steady state rather than to a time-varying efficient benchmark. We depart from the simplest inflation-forecast rule of Christiano et al. (2008) by responding to current rather than expected inflation and by including an output term; interest-rate smoothing and a response to credit growth are natural extensions that we do not pursue here. Choosing $\phi_\pi > 1$ satisfies the Taylor principle, but determinacy is a property of the full system and is verified numerically at the calibration used below.

The distinction between the policy target and the sticky nominal variable is substantive. Stabilising wage inflation would move the economy toward its flexible-wage allocation (Erceg et al., 2000). By targeting price inflation instead, rule (32) leaves room for wage rigidity to affect real allocations.

3.5 Exogenous Processes

Technology and news. Aggregate productivity follows

$$\log z_t = \rho_z \log z_{t-1} + \varepsilon_t^0 + D_t, \qquad \varepsilon_t^0 \sim N(0, \sigma_0^2), \tag{33}$$

where D_t is the delivery of previously announced news. At most one announcement is pending. When no announcement is pending, news arrives with probability λ_{news}. Its magnitude $a \sim N(0, \sigma_0^2)$ and horizon $H \sim 8 + \text{Binomial}(8, p_H)$ are drawn once and observed on arrival. News magnitudes and unanticipated productivity innovations therefore share the variance σ_0^2. The news state consists of its remaining horizon χ_t and magnitude a_t. A zero countdown means that no news is pending. At maturity, an independent confirmation draw F_t determines delivery:

$$D_t = \mathbf{1}\{\chi_{t-1} = 1\} F_t a_{t-1}, \qquad F_t \sim \text{Bernoulli}(\pi_{\text{confirm}}). \tag{34}$$

Agents know the confirmation probability, but not the outcome before maturity. After resolution, a new announcement may arrive in the same period. Otherwise, the magnitude is retained and the countdown falls by one. Appendix B.8 states the transition and timing convention.

This process retains the distinction between anticipated and unanticipated productivity movements in Christiano et al. (2008, 2010), but replaces certain delivery at a fixed horizon with sparse arrivals and probabilistic confirmation. The arrival rate, horizon distribution and confirmation probability are estimated rather than fixing the reversal date.

The unfulfilled-news experiment. The boom–bust experiment introduces positive news with magnitude $\bar{a}$ and horizon H at $t = 1$ and conditions on denial at $t = 1 + H$. During anticipation, agents assign probability π_{confirm} to delivery. At maturity, denial sets $D_{1+H} = 0$ and removes the pending announcement. No offsetting productivity surprise is required. With all other technology innovations set to zero and no further announcements, productivity remains constant. The fulfilled-news benchmark conditions on confirmation of the same announcement. Its anticipation path is identical because agents cannot know which outcome will occur.

The empirical credit-spread shock is closely associated with credit sentiment and generates a boom followed by a bust. This motivates the news experiment, in line with the credit-market interpretation of López-Salido et al. (2017). Optimism and pessimism here describe equilibrium responses to news and its denial; they are not separate exogenous belief shocks or departures from rational expectations.

Credit conditions and macro uncertainty. The enforcement parameter follows a level autoregression whose innovation variance is itself stochastic,

$$\zeta_t - \zeta_{ss} = \rho_\zeta \left(\zeta_{t-1} - \zeta_{ss}\right) + \exp(h_t/2)\, \varepsilon_{\zeta,t}, \tag{35}$$

$$h_t = (1-\rho_h)\bar{h} + \rho_h h_{t-1} + \sigma_h\, \varepsilon_{h,t}, \tag{36}$$

The untruncated specification follows Chatterjee et al. (2025). To keep enforcement positive in the global solution, define $\bar{\zeta}_t = (1-\rho_\zeta)\zeta_{ss} + \rho_\zeta \zeta_{t-1}$ and draw $\varepsilon_{\zeta,t}$ from a standard normal conditional on $\varepsilon_{\zeta,t} > -\bar{\zeta}_t / \exp(h_t/2)$. The volatility innovation satisfies $\varepsilon_{h,t} \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. The two innovations are generated from independent underlying draws, also independent of the technology and news draws. The unconditional mean of h_t is $\bar{h}$, so $\exp(\bar{h}/2)$ is the baseline scale of the enforcement innovation. An innovation to h_t changes that scale without changing the autoregressive location $\bar{\zeta}_t$. Positive-support truncation nevertheless raises the conditional mean above $\bar{\zeta}_t$ by a scale-dependent amount. We therefore interpret $\varepsilon_{h,t}$ as a macro-uncertainty shock, while retaining this truncation effect in both expectations and simulations. Uncertainty in the enforcement parameter represents macro uncertainty, measured empirically using the index of Jurado et al. (2015).

3.6 Equilibrium and Solution

A symmetric competitive equilibrium is a set of processes for the quantities, prices and multipliers

$$\{c_t, n_t, y_t, I_t, k_{t+1}, d_t, B_{t+1}\}, \qquad \{w_t, \Pi_t, \Pi_{w,t}, i_t, Q_t\}, \qquad \{\Lambda_t, \lambda_t, q_t, \mu_t\},$$

such that households and firms optimise, the enforcement constraint holds with complementary slackness, the policy rule (32) is satisfied, and markets clear, given the exogenous processes (33)–(36) and an initial state.

Goods market clearing is not an independent restriction. Substituting the tax rule and (23) into the household budget constraint (14), all nominal debt flows and the financing subsidy cancel and Walras' law delivers

$$y_t = c_t + I_t + \kappa \left(d_t - \bar{d}\right)^2 + \Theta_{w,t} w_t n_t. \tag{37}$$

The cancellation of nominal terms in (37) is an accounting result and does not make the nominal block irrelevant: inflation still revalues outstanding debt in (23), and the policy rate still enters the pledgeability of collateral in (24) and the cost of debt in (30).

We approximate the state-dependent decision rules with a DEQN (Azinovic et al., 2022), following the implementation described by Scheidegger (2026, Chapters 2 and 10). The state is $s_t = (k_t, b_t, c_{t-1}, w_{t-1}, \log z_t, \zeta_t, h_t, \chi_t, a_t)$, where $b_t = B_t/P_{t-1}$, so current repayment is b_t/Π_t. The network returns six coordinates for hours, the real wage, inflation, household marginal utility,

total payout outlay, and collateral slackness. The remaining allocation is recovered from the model identities. In particular, a signed coordinate ψ_t determines

$$S_t = \bar{S}\max\{\psi_t, 0\}, \qquad \mu_t = \bar{\mu}\max\{-\psi_t, 0\}, \tag{38}$$

with positive numerical scales $\bar{S}$ and $\bar{\mu}$. Thus (25) holds exactly, and the network can switch between regimes without imposing a binding calendar. Proposition 1 applies only to the deterministic steady state.

Training minimises the scaled residuals of internal habit, the household Euler equation, wage setting, labor demand, and the capital and debt Euler equations. Expectations use the same network at endogenous next-period states, with discrete news outcomes enumerated and continuous innovations integrated by quadrature. Perturbation and DynareOBC calculations serve as starting calculations and benchmarks; final estimation uses the DEQN decision rules, not a pruned third-order state-space system. Appendix B.9 gives the numerical procedure.

3.7 Inspecting the Mechanism

Three results organise the mechanism. The tax advantage of debt makes the enforcement constraint bind at the deterministic steady state. The investment adjustment cost makes collateral value vary with the investment rate. The payout adjustment cost allows that variation to affect the financing wedge. We establish these results first and then use them to trace the two experiments.

When does the constraint bind? At the deterministic steady state the payout equals its target, so $\lambda_{ss} = 1$, and the investment rate equals δ, so $q_{ss} = 1$ as well. The following result is the counterpart of Proposition 1 in Jermann and Quadrini (2012).

Proposition 1. *At the deterministic steady state with inflation target $\Pi^* = 1$, the multiplier on the enforcement constraint is strictly positive if and only if $\tau > 0$. Hence the constraint is strictly binding whenever $\tau > 0$, and the steady-state financing wedge is*

$$\frac{\mu_{ss}}{\lambda_{ss}} = \frac{\tau(1-\beta)}{\zeta_{ss}\,(1-\tau+\tau\beta)}. \tag{39}$$

Proof. With $\Pi^* = 1$, (18) gives $1 + i_{ss} = 1/\beta$, so $R_{ss} = 1 + (1-\tau)(1-\beta)/\beta = (1-\tau+\tau\beta)/\beta$ and $\beta R_{ss} = 1 - \tau + \tau\beta$. The debt condition (30) is a Kuhn–Tucker condition and holds whether or not the constraint binds, with $\mu_{ss} \geq 0$ and $\mu_{ss} = 0$ if the constraint is slack. Evaluated at the steady state with $\lambda_{ss} = 1$ it reads $\beta = 1/R_{ss} - \mu_{ss}\zeta_{ss}\beta$, so that

$$\mu_{ss}\,\zeta_{ss}\,\beta R_{ss} = 1 - \beta R_{ss} = \tau(1-\beta).$$

If $\tau > 0$ the right-hand side is strictly positive, so $\mu_{ss} > 0$ and the constraint must bind.

Conversely, $\tau = 0$ implies $\mu_{ss} = 0$. In that case complementary slackness does not determine whether the constraint holds with equality or is slack; the answer depends on the equilibrium selection for debt. Solving for μ_{ss} delivers (39). □

The tax advantage of debt is therefore the reason the constraint binds in steady state: firms lever up to exhaust the shield until enforcement stops them. Equation (39) also shows that the average wedge is decreasing in ζ_{ss}, so the steady-state severity of the friction is disciplined by a single parameter, and it is increasing in τ and in the steady-state real rate.

Binding in steady state is necessary but not sufficient for credit conditions to matter. The next result identifies the case in which they do not.

Proposition 2. *If $\tau = 0$, $\kappa = 0$, and $\zeta_t > 0$, then $\mu_t = 0$ at every date. Innovations to ζ_t, whether to its level or its volatility, have no effect on real allocations or on the associated price system. Debt and payout remain financing variables whose values can depend on equilibrium selection.*

Proof. With $\kappa = 0$ the payout condition gives $\lambda_t \equiv 1$, so (30) reads $\mathbb{E}_t[m_{t,t+1}/\Pi_{t+1}] = 1/R_t - \mu_t\zeta_t/(1+i_t)$. The bond pricing condition (18) gives $\mathbb{E}_t[m_{t,t+1}/\Pi_{t+1}] = 1/(1+i_t)$, and $\tau = 0$ implies $R_t = 1 + i_t$. Combining, $\mu_t\zeta_t = 0$, so $\mu_t = 0$ for $\zeta_t > 0$. With $\mu_t = 0$ the parameter ζ_t drops out of the conditions that determine real allocations and prices. It remains in the enforcement constraint, which restricts the admissible combinations of debt and payout. Because payouts are costless when $\kappa = 0$, those financing choices need not be unique and do not feed back into the real allocation. □

Proposition 2 is the analogue of the neutrality result in Jermann and Quadrini (2012), and it makes clear that the financing block is not a free amplification device: with frictionless payouts and no tax shield, the real side of the model collapses to a sticky-wage economy with flexible prices. Credit conditions may select a balance-sheet configuration, but they do not alter real allocations or prices.

Remark 1. The same algebra with $\kappa = 0$ but $\tau > 0$ yields $\mu_t = \tau i_t/(\zeta_t R_t)$ at every date. The multiplier is then pinned down statically by the policy rate and the enforcement parameter, and is *invariant to the value of collateral* Q_t. Credit conditions still move the labor wedge through ζ_t, but a revaluation of pledged capital does not. A positive payout adjustment cost is therefore necessary for the collateral-price channel studied here: it makes λ_t endogenous and time-varying, and it is only through the path of λ_t that the value of collateral can reach the wedge, as (41) below makes explicit.

The collateral price is an investment-rate object. Combining the investment condition with the valuation rule (26) eliminates both multipliers and leaves a single relationship.

Lemma 1. *On any equilibrium path with* $\lambda_t > 0$ *and* $1 - \phi\left(I_t/k_t - \delta\right) > 0$,

$$Q_t = \left[1 - \phi\left(\frac{I_t}{k_t} - \delta\right)\right]^{-1}, \tag{40}$$

so that $Q_{ss} = 1$ *and* $\partial Q_t / \partial (I_t/k_t) = \phi\, Q_t^2 > 0$ *whenever* $\phi > 0$. *If* $\phi = 0$ *then* $Q_t \equiv 1$.

Lemma 1 makes collateral value a strictly increasing function of the investment rate. Higher desired investment therefore raises the price applied to a predetermined collateral stock and expands borrowing capacity. If $\phi = 0$, Q_t is constant and this revaluation channel disappears. Combined with Remark 1, the result assigns distinct roles to the two adjustment costs: ϕ makes pledgeable value move, while κ transmits that movement to the financing wedge.

What pins down the financing wedge. Lemma 1 describes how the value of collateral moves; it does not by itself say how the labor wedge responds. Rearranging (30),

$$\frac{\mu_t}{\lambda_t} = \frac{1 + i_t}{\zeta_t}\left[\frac{1}{R_t} - \frac{\mathbb{E}_t[m_{t,t+1}\lambda_{t+1}/\Pi_{t+1}]}{\lambda_t}\right]. \tag{41}$$

The collateral price does not appear in (41). Conditional on the policy rate, enforcement, and the expected path of the marginal value of internal funds, the debt margin determines the wedge. If the constraint binds, (24) determines debt residually, $b_{t+1} = (1 + i_t)\left[Q_t k_t - w_t n_t/\zeta_t\right]$ with $b_{t+1} \equiv B_{t+1}/P_t$. A revaluation of collateral therefore reaches the wedge only indirectly. It changes how much the firm can borrow; the additional borrowing is reconciled with investment and payouts in the flow of funds; and the resulting movement in λ_t relative to its expected future value moves the wedge through (41). The wedge falls when the marginal value of internal funds is expected to rise, when the firm is flush today relative to tomorrow, and rises in the opposite case. Because μ_t and λ_t both respond, the direction of the movement is not signed by the structure of the model; it is a general-equilibrium outcome. When $\kappa = 0$, $\lambda_t \equiv 1$ and (18) reduces (41) to the expression in Remark 1, which is why the payout friction is necessary for this route to operate.

In a slack state, debt instead satisfies $b_{t+1} < (1 + i_t)[Q_t k_t - w_t n_t/\zeta_t]$ and $\mu_t = 0$. The contemporaneous financing wedge vanishes, and labor demand reduces to $w_t = (1 - \theta) y_t / n_t$. Borrowing is then determined jointly by the flow of funds and the intertemporal conditions, not by exhausting collateral. Future binding states can still affect current choices through the Euler equations. Equation (41) holds in both regimes; its right-hand side is zero in a slack equilibrium.

How unfulfilled news can generate a boom and then a bust. Consider positive news at $t = 1$ with maturity at $t = 1 + H$. Its possible delivery raises the expected marginal product of capital and, because capital also serves as collateral, it can raise the expected collateral payoff $\mu_{t+1}\zeta_{t+1}Q_{t+1}$ in (29). To the extent that these raise q_t relative to λ_t, the collateral price rises and, by Lemma 1, so does the investment rate. These are two expressions of the same equilibrium

movement. The higher collateral price raises borrowing capacity against a predetermined capital stock. The firm's use of that capacity and its allocation between investment and payouts determine the path of λ_t. Through (41) that path moves the financing wedge, and through (28) the wedge moves labor demand. A boom arises if the wedge falls, so that employment and output rise before any productivity gain has materialised. If capacity expands sufficiently, the constraint becomes slack and the wedge reaches zero. Further movements in hours then reflect wages, productive capacity and intertemporal choices rather than a further decline in the financing wedge.

If news is denied at $t = 1 + H$, the expected productivity gain is removed. Investment and collateral value can fall while the debt accumulated during anticipation remains due. A lower value of collateral against that debt can make the constraint bind again, restoring the hiring wedge and producing a contraction. This is the model's representation of a reversal in credit-market sentiment: anticipated gains support borrowing, and disappointment leaves the economy with the resulting repayment obligations. Denial contributes no productivity innovation. The boom–bust response, the duration of slackness, and rebinding after denial are quantitative outcomes to be checked in the DEQN solution, not restrictions imposed on the decision rules.

Why nominal wage rigidity is required. The need for wage rigidity arises from the news experiment. Following an uncertainty shock, a higher financing wedge can generate a contraction in output, consumption, investment and hours even with flexible wages. During news anticipation, however, a lower wedge is insufficient to produce an expansion in all four variables. In the flexible-wage benchmark, consumption rises while output, investment and hours fall. The obstacle resembles the co-movement problem in the real model discussed by Basu and Bundick (2017), although here it arises in response to news.

When $\phi_w = 0$, (19) requires $w_t = \mu_w \, \alpha n_t^{\nu} / \Lambda_t$. Anticipated productivity gains raise consumption and reduce the household's marginal value of wealth, increasing the wage required to supply a given amount of labor. This wage increase offsets the expansionary effect of the lower financing wedge in (28), so hours fall despite easier credit. With $\phi_w > 0$, the real wage need not satisfy the static markup condition each period. Slower nominal wage adjustment limits the wage response during anticipation, allowing labor demand and hours to rise alongside consumption, output and investment. Wage rigidity is therefore introduced to obtain co-movement in the news-driven expansion; it is not needed solely to obtain co-movement in the uncertainty-driven contraction.

The policy rule then determines how much of that shift survives. Through (31), a boom in which the real wage rises faster than nominal wages adjust is a boom with falling price inflation, and (32) responds by cutting the policy rate. This lowers the effective cost of debt in (22). It may also raise the present value of the collateral payoff in (29) by lowering the real discount rate, although the firm discounts with the household's stochastic discount factor rather than directly with i_t. The output term pushes in the opposite direction. The net response depends on ϕ_π relative to ϕ_y. Christiano et al. (2008) find, in a related setting, that the boom following

an unfulfilled signal is substantially larger than the efficient response to the same signal, and attribute the difference to a policy rule that fails to track the rise in the natural rate. Section 3.8 estimates ϕ_π and ϕ_y rather than fixing them, precisely because they are part of the mechanism.

Macro uncertainty. The second experiment raises the scale of enforcement innovations through $\varepsilon_{h,t}$ in (36), retaining the positive-support truncation described above. In the debt margin (41), greater dispersion of future enforcement changes the expected path of the marginal value of internal funds, because firms anticipate states in which borrowing capacity contracts and payouts must be cut. In the capital Euler equation (29) it changes the expected collateral payoff $\mathbb{E}_t[\mu_{t+1}\zeta_{t+1}Q_{t+1}]$, which depends on the covariance between future enforcement, the multiplier and the collateral price as well as on their means. In the environment of Chatterjee et al. (2025) the net effect is a precautionary tightening in which output, consumption, investment and hours fall together. Here the same forces operate through (41) and (29), and any fall in investment lowers the collateral price through Lemma 1 and contracts borrowing capacity further. Unlike the anticipation experiment, precautionary tightening can raise a positive multiplier without a switch to slackness. Whether the constraint remains binding throughout this response must be checked in the solution. Both experiments act through the financing margin, but they need not generate the same temporal pattern.

This distinction corresponds to the empirical separation of a predominantly contractionary uncertainty shock from a credit-spread shock with boom–bust dynamics. The model is asked to reproduce both responses with one financing block. It does not identify the two shocks by the sign of the spread response alone. Because this environment adds payout costs, nominal debt and a monetary block to that of Chatterjee et al. (2025), neither four-way co-movement nor a subsequent reversal follows analytically. These are restrictions against which the DEQN responses are evaluated.

3.8 Calibration and Estimation

Following Christiano et al. (2005), we split the parameters into two groups. The first is fixed ex ante, either from steady-state relationships or from values that are standard in the literature. The second is estimated by minimising the distance between the model's impulse responses, computed from the DEQN solution, and the two sets of empirical responses identified in Section 2: the responses to a credit shock, which the unfulfilled-news experiment is asked to match, and the responses to an uncertainty shock, which the volatility experiment is asked to match. Table 1 lists both groups. Its numerical values are benchmark calibrations or starting values, not reported estimates.

For estimation, the estimated parameters enter the network as additional inputs, or pseudo-states, as in Scheidegger (2026, Chapters 9–10). The network is trained over their admissible ranges with each parameter vector held fixed along a simulated path. This approximates a family

of equilibrium decision rules $g(s_t;\vartheta)$ without re-solving the model at every objective evaluation. For a candidate ϑ, stack the model generalised impulse responses over forty quarters into $m(\vartheta)$ and the corresponding empirical responses into $\hat{m}$. The estimate minimises

$$J(\vartheta) = [m(\vartheta) - \hat{m}]' W [m(\vartheta) - \hat{m}], \tag{42}$$

where W is constructed from the covariance matrix of the empirical responses. Common random numbers are retained across candidates, and the search uses derivative-free methods because the news outcomes are discrete.

The credit-spread target is the empirical shock identified after separating the uncertainty component; its model counterpart is positive news followed by denial, not an enforcement-level innovation. The uncertainty target is matched to an innovation in h_t. Output, consumption, investment, hours, the policy rate and inflation provide direct model counterparts. The empirical credit spread of Gilchrist and Zakrajšek (2012), excess bond premia and credit sentiment motivate the financing interpretation, but this model has no default-risk or bond-pricing block that reproduces those series directly. They are therefore not equated to μ_t/λ_t or included as model observables without an additional measurement relation.

The calibrated block is standard. The discount factor implies an annualised steady-state real rate of about four percent; the capital share, depreciation rate and technology persistence take conventional values. The elasticity of substitution across labor varieties implies a steady-state wage markup of twenty percent, and the tax parameter follows Jermann and Quadrini (2012). News horizons range from eight to sixteen quarters, with mean $8 + 8p_H$. Their distribution is estimated using the timing of the empirical credit-shock response. Fulfilled and denied news experiments use the same magnitude and horizon for comparability.

The data discipline the two parameters that govern the collateral-price channel. Lemma 1 makes ϕ the elasticity of collateral value to the investment rate, while Remark 1 shows that κ controls whether collateral revaluation reaches the labor wedge. The estimate of ϕ_w is reported together with the contract duration implied by the mapping in Born and Pfeifer (2020).

The steady-state enforcement parameter admits a similar external check. Appendix B.5 shows that, holding the calibrated block fixed, the deterministic steady state maps ζ_{ss} monotonically into the ratio of corporate debt to quarterly output, so the estimate can be read against observed leverage even though it is not calibrated to it. We report that implied ratio alongside the estimate, together with the two admissibility restrictions derived in the same appendix.

Every candidate parameter vector must satisfy two restrictions. First, the deterministic steady state must exist, with positive intertemporal debt and a binding enforcement constraint. At the values in Table 1, this requires ζ_{ss} to exceed approximately 0.06; there is no upper bound. Appendix B.5 derives the restrictions and explains why the direction of the bound differs from Chatterjee et al. (2025). Second, the DEQN must satisfy the equilibrium residual, feasibility and stability checks in Appendix B.9 throughout the parameter region used for estimation.

Complementarity is imposed by (38); both regimes are admissible. Positive consumption surplus, capital, prices and marginal values, and the investment branch in (B.25), are checked separately. Enforcement remains positive under the same truncated transition in expectations and simulations. Failed paths and domain exits are reported, not hidden by averaging the surviving paths. At the estimate, a dedicated fixed-parameter network is solved with stricter accuracy requirements and its responses are compared with those of the parameterised network. Artificial-data parameter recovery and local identification checks assess the estimator.

Table 1: Parameters

Parameter	Description	Value	Treatment
Preferences and technology			
β	Discount factor	0.9900	Calibrated
σ	Consumption curvature	4.9831	Estimated
ι	Habit persistence	0.5318	Estimated
ν	Inverse Frisch elasticity	0.3903	Estimated
θ	Capital share	0.3300	Calibrated
δ	Depreciation rate	0.0250	Calibrated
ϕ	Investment adjustment cost	12.8704	Estimated
labor market and nominal rigidity			
η_w	Elasticity of substitution across varieties	9.4619	Estimated
ϕ_w	Rotemberg wage adjustment cost	63.9835	Estimated
Financial structure			
κ	Payout adjustment cost	0.0917	Estimated
τ	Tax advantage of debt	0.350	Calibrated
ζ_{ss}	Steady-state enforcement parameter	0.1516	Estimated
Monetary policy			
ϕ_π	Response to inflation	2.1254	Estimated
ϕ_y	Response to output	0.2500	Calibrated
Exogenous processes			
ρ_z	Persistence of technology	0.950	Calibrated
σ_0	S.d. of productivity surprises and news	0.0080	Estimated
λ_{news}	News arrival probability	0.0312	Estimated
p_H	News horizon distribution parameter	0.7480	Estimated
π_{confirm}	News confirmation probability	0.9354	Estimated
ρ_ζ	Persistence of enforcement parameter	0.950	Calibrated
$\bar{h}$	Mean log variance of enforcement innovation	-10.27	Calibrated
ρ_h	Persistence of log volatility	0.8836	Estimated
σ_h	S.d. of volatility shock	0.1378	Estimated

Notes. The model is calibrated at quarterly frequency. Calibrated parameters are fixed prior to estimation; estimated parameters are chosen to minimise the distance between model-implied and empirical impulse responses. Values shown for estimated parameters are starting values; dashes indicate news-process parameters for which no final values are reported here. The value of $\bar{h}$ is the posterior mean reported for the baseline stochastic-volatility specification in Chatterjee et al. (2025).

3.9 Model Results

Figures 10 and 11 compare the responses to macro uncertainty and denied productivity news with their empirical targets. The comparison concerns two restrictions: co-movement among output, consumption, investment and hours, and the distinction between a contraction followed mainly by recovery and an expansion followed by a contraction. The latter is the feature that distinguishes the empirical credit-spread shock after separating the uncertainty component. Since these responses are estimation targets, their fit assesses the model's ability to account for the conditional dynamics rather than providing an independent test of the mechanism.

Macro uncertainty. Figure 10 shows that an increase in macro uncertainty lowers all four quantities. The responses are hump shaped: activity falls further after impact and then recovers gradually. Investment and hours respond most strongly, with approximate troughs of -1.6 and -0.65, respectively, in the figure's impact-normalised units. Consumption reaches a trough of about -0.3. Most of the decline occurs within the first ten quarters. Output, consumption and investment remain below baseline at the end of the forty-quarter window, while hours exhibit a small late overshoot. Thus the dominant response is a contraction and recovery, although the hours response is not entirely one sided.

The model follows the empirical decline and recovery in output and hours reasonably closely. Its investment contraction is somewhat larger, and its consumption recovery is faster than the empirical median. There is a more substantial difference in the uncertainty panel. The model's uncertainty response decays toward zero from above, whereas the empirical uncertainty response becomes negative before returning toward baseline. The figure therefore supports the contractionary quantity responses more closely than the subsequent path of the uncertainty measure itself.

The financing mechanism differs from the demand-uncertainty channel of Basu and Bundick (2017). Here greater dispersion in enforcement exposes firms to future shortages of borrowing capacity and changes the value of internal funds. A higher financing wedge shifts labor demand inward through (28), a hiring channel related to that of Jermann and Quadrini (2012). As discussed in Section 3.7, this channel can generate four-way contraction even with flexible wages. The quantity responses are consistent with that mechanism, but the figure alone does not establish the multiplier's path or whether the ICC remains binding at every date.

Unfulfilled productivity news. Figure 11 displays a different temporal pattern. Positive news produces an initial expansion in output, consumption, investment and hours, followed by responses below baseline. The expansion peaks within roughly the first ten quarters. Output rises by about 0.3 percent, consumption by 0.25 percent, investment by 1.2 percent and hours by 0.6 percent. Investment and hours subsequently turn negative, followed by output and consumption. The contraction is persistent, with the largest negative responses occurring broadly between

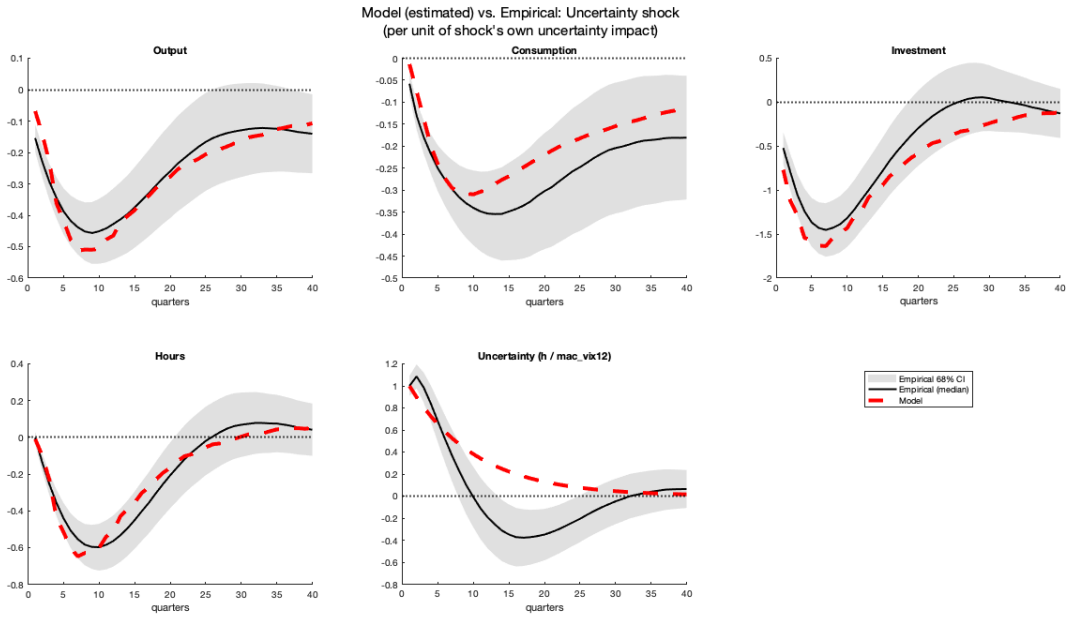


Figure 10: Responses to a macro-uncertainty shock

Notes. Red dashed lines show the supplied model responses, black lines the empirical medians, and shaded areas the empirical 68 percent confidence intervals. Each set of responses is normalised by the impact response of uncertainty in the corresponding model or empirical system. The horizon is forty quarters. Shading describes empirical uncertainty, not uncertainty about the model's parameters or numerical solution.

quarters twenty-five and thirty-five. All four quantities remain below baseline at quarter forty.

The model captures the empirical ordering of expansion and contraction, including the relatively large movements in investment and hours. The fit is less close for consumption: its model expansion lasts longer and its subsequent contraction is larger than the empirical median. The model also generates deeper and more persistent declines in output, investment and hours. These differences matter because the empirical consumption cycle is weaker than the cycles in the other quantities. Broad empirical confidence bands do not by themselves establish that the discrepancies are quantitatively unimportant.

The expansion is consistent with the collateral-revaluation channel in Lemma 1. Anticipated productivity gains encourage investment and raise the value attached to predetermined capital. Easier financing can reduce the hiring wedge, while slow nominal wage adjustment allows hours to rise alongside consumption. After denial, the anticipated gain is removed, but the financing and investment choices made during anticipation carry forward. A decline in collateral value against inherited debt can restore a binding constraint and reverse the expansion. This interpretation requires the debt, slack and multiplier paths described in Section 3.7; the four quantity panels cannot independently verify the proposed switch from binding to slack and back to binding.

The fifth panel shows model-implied Tobin's Q. Its initial rise and subsequent fall below

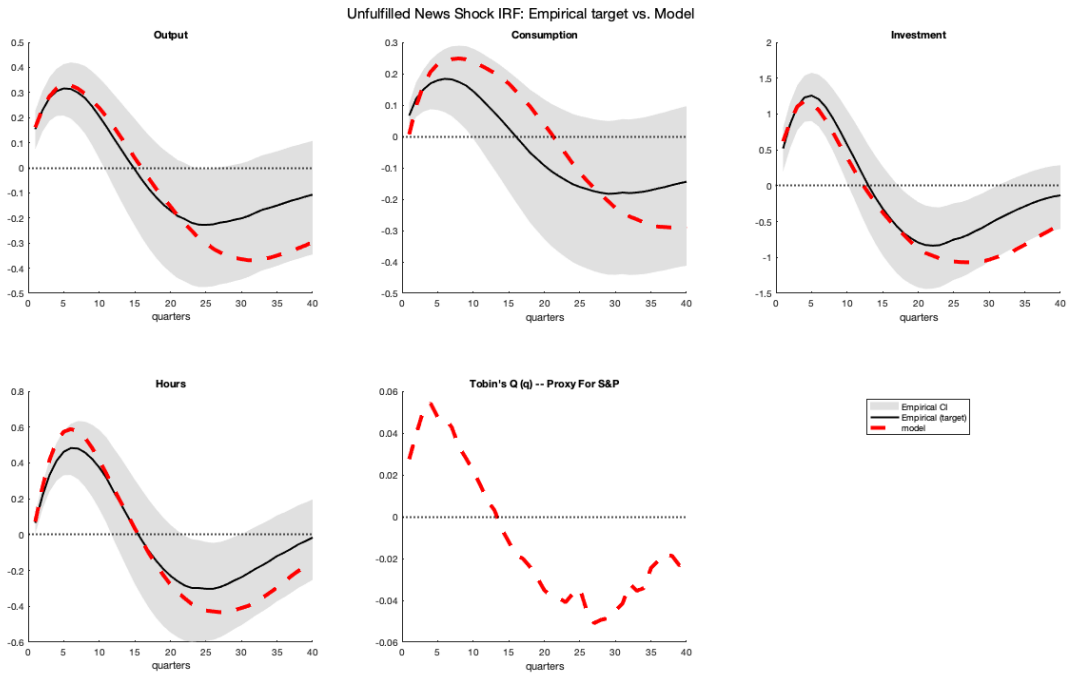


Figure 11: Responses to positive productivity news followed by denial

Notes. Red dashed lines show the supplied model responses, black lines the empirical credit-shock targets, and shaded areas the empirical 68 percent confidence intervals. Quantity responses are percentage deviations from baseline over forty quarters. The fifth panel shows model-implied Tobin's Q, for which there is no exact counterpart in the empirical IRFs. Its boom–bust pattern is compared qualitatively with the empirical S&P response; no empirical confidence band is plotted for Q.

baseline accompany the expansion and contraction in real activity. This valuation cycle is consistent with the link between investment and collateral value in Lemma 1. It also parallels the boom–bust response of S&P prices to the empirical credit-spread shock. Tobin's Q values installed capital in units of goods, whereas the S&P index measures equity prices. The comparison therefore concerns the direction and sequence of valuation movements, not an exact observable counterpart or a quantitative match of their magnitudes.

Connection to the empirical credit cycle. The empirical credit-spread shock produces boom–bust responses in real activity and stock prices, accounts for a substantial share of excess bond premia fluctuations, and is closely associated with credit sentiment. These findings motivate interpreting anticipation and denial as optimism and its reversal, consistent with the sequence of buoyant credit conditions and subsequent weakness emphasised by López-Salido et al. (2017). In this model, that sequence results from rational responses to a pending announcement and its resolution. The model does not reproduce their investor-belief or risk-appetite mechanism, nor does it directly generate excess bond premia, the high-yield issuance ratio or the S&P index.

Nevertheless, the boom–bust response of model-implied Q provides a valuation analogue to the empirical stock-price cycle alongside the co-movement in real activity.

The two experiments account for distinct conditional patterns within the same financing environment. Uncertainty mainly generates contraction and recovery; denied news generates an expansion and a subsequent contraction. This distinction is consistent with the empirical finding that separating uncertainty reveals a credit-related cycle that is less apparent in the jointly identified shock. It does not establish the relative contributions of the two model shocks to aggregate volatility or their spectral densities. Those implications require simulated shock contributions and spectra in addition to the impulse responses reported here.

4 Conclusion

Hours worked has a pronounced spectral peak and sign-changing autocorrelation, yet the shock that explains most of its business-cycle variation produces no clear bust. We show that the MBC shock masks two disturbances. Higher macro uncertainty generates a broad recession followed by recovery. An orthogonal credit sentiment shock instead raises output, consumption, investment, hours, and stock prices before these variables reverse. It also explains substantial variation in the excess bond premium and is closely related to the high-yield issuance share. The uncertainty component dominates within the MBC shock, muting the reversal in its impulse response even though it remains visible in the spectrum.

These facts discipline a model in which firms borrow against collateral to finance their wage bill. Anticipated productivity news raises investment and collateral value and relaxes the borrowing constraint. If the news is denied, inherited debt and lower collateral value tighten financing conditions and reverse the expansion. Uncertainty about enforcement operates through the same financial environment but immediately raises the financing wedge and contracts activity. Slow nominal wage adjustment generates comovement after news, while the financial wedge is sufficient for the contractionary uncertainty response. A global neural-network solution allows the borrowing constraint to switch endogenously between binding and slack regimes.

The quantitative model reproduces the central conditional distinction and a boom–bust movement in Tobin's Q consistent with the stock-price response. The exercise also has limits: the empirical responses are estimation targets, the model does not directly generate the excess bond premium or the high-yield issuance share, and some quantity responses are too persistent. Matching unconditional variance shares and shock-specific spectra is the natural next test of how much business-cycle variation can be attributed to uncertainty and credit-market reversals.

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Appendix A: Data Construction and Empirical Implementation

This appendix records the construction of the variables used in Section 2. The objective is to make the empirical inputs reproducible while keeping measurement details out of the main text.

A.1. Baseline VAR variables

Table A.1: Construction of the baseline variables

Variable	Construction	Source
Output	100 log of real GDP divided by the civilian noninstitutional population aged sixteen and older.	BEA National Income and Product Accounts; BLS population series.
Consumption	100 log of real consumption of nondurable goods and services per capita. Nominal expenditure shares are used to allocate real GDP to these two components.	BEA National Income and Product Accounts.
Investment	100 log of real gross private domestic investment plus consumer durable expenditure, divided by population. Nominal expenditure shares are used to form the aggregate.	BEA National Income and Product Accounts.
Hours	100 log of total hours in the nonfarm business sector divided by the civilian noninstitutional population aged sixteen and older.	BLS nonfarm business hours and population series.
Interest rate	Quarterly average of the effective federal funds rate, divided by four so that the rate is expressed at a quarterly frequency.	Federal Reserve Board.
TFP	Cumulated quarterly growth in utilization-adjusted total factor productivity. The reported annualized growth rate is divided by four before cumulation.	Fernald (2014).
Credit spread	Quarter-end Gilchrist–Zakrajšek corporate bond spread, divided by four.	Gilchrist and Zakrajšek (2012).
Macro uncertainty	Quarterly average of the monthly twelve-month-ahead macro uncertainty index, expressed as 100 log of the index.	Jurado et al. (2015).

The four real aggregates follow the construction used by Angeletos et al. (2020). The baseline

estimation begins in 1973:I and ends in 2024:IV. The current estimation input removes 2020:II and 2020:III, the two observations most affected by the abrupt pandemic shutdown and reopening. Apart from these exclusions, quarterly observations are aligned by calendar date before the sample is selected.

A.2. Comparison and auxiliary variables

The specification used for comparison with Angeletos et al. (2020) adds labor productivity and the labor share. Labor productivity is 100 log of nonfarm business output per hour, and the labor share is 100 log of the BLS nonfarm business labor-share index. Appendix specifications also consider inflation, measured as $100 \log(P_t/P_{t-1})$ using the GDP implicit price deflator. Inflation is not part of the eight-variable baseline VAR.

The auxiliary financial series are constructed separately because their samples differ from that of the baseline VAR. The real per-capita S&P index is $100 \log[S_t/(0.01 P_t N_t)]$, where S_t is the quarter-end S&P price index, P_t is the GDP deflator, and N_t is population. The excess bond premium is the quarter-end series of Gilchrist and Zakrajšek (2012), divided by four. The high-yield ratio is the share of nonfinancial corporate bond issuance rated high yield, constructed by Greenwood and Hanson (2013); the available quarterly series runs from 1983:I through 2016:I. These variables are projected on the contemporaneous baseline VAR vector and two of its lags, as described in equations (8) and (9).

A.3. Prior and posterior computation

The VAR contains two lags and an intercept. The Minnesota prior centers the first own lag of every variable at one. Its four hyperparameters are $(0.2, 0.5, 2, 10^5)$: the first controls overall tightness, the second shrinks cross-variable lags relative to own lags, the third increases shrinkage with lag length, and the fourth makes the intercept effectively diffuse. Innovation variances from univariate AR(1) regressions scale the prior across variables. The covariance matrix has an inverse-Wishart prior. The Gibbs sampler uses 50,000 draws, discards the first 49,000, and retains the remaining 1,000 draws for posterior summaries. Each draw is checked for stability before its frequency-domain rotation and impulse responses are computed.

Appendix B: Derivations of the Model

This appendix derives every optimality condition reported in Section 3, characterises the deterministic steady state in closed form, proves the results stated in Section 3.7, lists the complete equilibrium system, and describes how the news experiment and the impulse responses are computed. Throughout, φ_t denotes the multiplier on the production function and should not be confused with the investment adjustment cost parameter ϕ or the wage adjustment cost parameter ϕ_w. Table B.1 at the end of the appendix collects all notation.

B.1 The Household's Problem

Setup. Substituting the labor demand curve (15) directly into both the objective (13) and the budget constraint (14), and attaching the multiplier $\beta^t \Lambda_t(j)$ to the date-t budget constraint, household j chooses $\{c_t(j), B_{t+1}(j), W_t(j)\}_{t\geq 0}$ to maximise

$$\begin{aligned}\mathcal{L}_0 = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Bigg\{ & \frac{(c_t(j) - \iota c_{t-1}(j))^{1-\sigma} - 1}{1-\sigma} \\ & - \frac{\alpha}{1+\nu} \left[\left(\frac{W_t(j)}{W_t} \right)^{-\eta_w} n_t \right]^{1+\nu} \\ & + \Lambda_t(j) \Bigg[\big(1 - \Theta_{w,t}(j)\big) \frac{W_t(j)}{P_t} \left(\frac{W_t(j)}{W_t} \right)^{-\eta_w} n_t \\ & \qquad + \frac{B_t(j)}{P_t} + d_t - T_t - c_t(j) - \frac{B_{t+1}(j)}{P_t(1+i_t)} \Bigg] \Bigg\}. \end{aligned} \tag{B.1}$$

The household takes W_t, n_t, P_t, i_t, d_t and T_t as given.

Consumption. Because habit is internal, $c_t(j)$ enters $\mathcal{L}_0$ at two dates: directly through date-t felicity, and indirectly through the habit stock $\iota c_t(j)$ that appears in date-$(t+1)$ felicity. Writing $X_t(j) \equiv (c_t(j) - \iota c_{t-1}(j))^{-\sigma}$ for the flow marginal felicity and noting that

$$\frac{\partial}{\partial c_t(j)} \left[\frac{(c_{t+1}(j) - \iota c_t(j))^{1-\sigma}}{1-\sigma} \right] = -\iota \left(c_{t+1}(j) - \iota c_t(j) \right)^{-\sigma} = -\iota X_{t+1}(j), \tag{B.2}$$

the derivative of (B.1) with respect to $c_t(j)$ collects $\beta^t X_t(j)$ from date t, $-\beta^{t+1} \iota \, \mathbb{E}_t X_{t+1}(j)$ from date $t+1$, and $-\beta^t \Lambda_t(j)$ from the budget constraint. Dividing by β^t and setting the result to zero gives the household-level condition

$$\Lambda_t(j) = X_t(j) - \iota \beta \, \mathbb{E}_t[X_{t+1}(j)] \, . \tag{B.3}$$

The correction term is strictly negative for $\iota > 0$, so $\Lambda_t(j) < X_t(j)$: the household values an extra unit of wealth less than the flow term alone would suggest, because consuming it raises tomorrow's reference stock. Under external habit the second term is absent and $\Lambda_t(j) = X_t(j)$. After symmetry is imposed, this condition becomes (17).

Bonds. Differentiating (B.1) with respect to $B_{t+1}(j)$ gives

$$-\beta^t \frac{\Lambda_t(j)}{P_t(1+i_t)} + \beta^{t+1}\,\mathbb{E}_t\left[\frac{\Lambda_{t+1}(j)}{P_{t+1}}\right] = 0 \quad \Longrightarrow \quad \frac{\Lambda_t(j)}{P_t(1+i_t)} = \beta\,\mathbb{E}_t\left[\frac{\Lambda_{t+1}(j)}{P_{t+1}}\right]. \tag{B.4}$$

Multiplying through by $P_t/\Lambda_t(j)$ and using $m_{t,t+1}(j) \equiv \beta\Lambda_{t+1}(j)/\Lambda_t(j)$ and $\Pi_{t+1} = P_{t+1}/P_t$ delivers the household-level bond-pricing condition

$$1 = (1+i_t)\,\mathbb{E}_t\left[\frac{m_{t,t+1}(j)}{\Pi_{t+1}}\right]. \tag{B.5}$$

After symmetry is imposed, this condition becomes (18). The discount factor $m_{t,t+1}(j)$ prices real payoffs; a nominal payoff must additionally be deflated, which is the source of the $1/\Pi_{t+1}$ term. The condition holds exactly under uncertainty, so no separate Fisher equation is imposed anywhere in the model.

The posted wage. The wage $W_t(j)$ enters (B.1) at date t through three channels and at date $t+1$ through one. We take them in turn, using $\partial n_t(j)/\partial W_t(j) = -\eta_w n_t(j)/W_t(j)$ throughout.

(i) labor disutility at t. Since a higher posted wage reduces the hours demanded of variety j,

$$\frac{\partial}{\partial W_t(j)}\left[-\frac{\alpha}{1+\nu}n_t(j)^{1+\nu}\right] = -\alpha n_t(j)^{\nu}\frac{\partial n_t(j)}{\partial W_t(j)} = +\alpha\eta_w\, n_t(j)^{\nu}\frac{n_t(j)}{W_t(j)}, \tag{B.6}$$

which is positive: raising the wage is attractive in part because it economises on work effort.

(ii) labor income at t. Using (15) we may write $W_t(j)n_t(j) = W_t(j)^{1-\eta_w}W_t^{\eta_w}n_t$, so that

$$\frac{\partial\left[W_t(j)n_t(j)\right]}{\partial W_t(j)} = (1-\eta_w)\,n_t(j), \tag{B.7}$$

which is negative because demand for the variety is elastic.

(iii) The adjustment cost at t. From (16) and $\Pi_{w,t}(j) = W_t(j)/W_{t-1}(j)$ with $W_{t-1}(j)$ predetermined,

$$\frac{\partial\Theta_{w,t}(j)}{\partial W_t(j)} = \frac{\phi_w\big(\Pi_{w,t}(j)-1\big)}{W_{t-1}(j)} = \frac{\phi_w\big(\Pi_{w,t}(j)-1\big)\Pi_{w,t}(j)}{W_t(j)}. \tag{B.8}$$

Combining (ii) and (iii), the date-t income term contributes

$$\Lambda_t(j)\frac{n_t(j)}{P_t}\Big[(1-\eta_w)\big(1-\Theta_{w,t}(j)\big) - \phi_w\big(\Pi_{w,t}(j)-1\big)\Pi_{w,t}(j)\Big]. \tag{B.9}$$

(iv) The adjustment cost at $t+1$. Since $\Pi_{w,t+1}(j) = W_{t+1}(j)/W_t(j)$ depends on the wage posted today,

$$\frac{\partial \Pi_{w,t+1}(j)}{\partial W_t(j)} = -\frac{\Pi_{w,t+1}(j)}{W_t(j)}, \qquad \frac{\partial \Theta_{w,t+1}(j)}{\partial W_t(j)} = -\frac{\phi_w\big(\Pi_{w,t+1}(j)-1\big)\Pi_{w,t+1}(j)}{W_t(j)}. \tag{B.10}$$

Because the cost enters the date-$(t+1)$ budget constraint multiplied by $W_{t+1}(j)n_{t+1}(j)/P_{t+1}$, and $W_{t+1}(j)/W_t(j) = \Pi_{w,t+1}(j)$, this term contributes

$$\beta\,\mathbb{E}_t\left[\Lambda_{t+1}(j)\phi_w\big(\Pi_{w,t+1}(j)-1\big)\Pi_{w,t+1}(j)^2\frac{n_{t+1}(j)}{P_{t+1}}\right]. \tag{B.11}$$

Note that a higher wage today lowers next period's wage inflation and therefore *reduces* next period's adjustment cost; this is the forward-looking term that makes wage setting dynamic.

Summing (i)–(iv), setting the total to zero, multiplying through by $P_t/\left(\Lambda_t(j)n_t(j)\right)$, and imposing symmetry ($W_t(j) = W_t$, $n_t(j) = n_t$, $\Theta_{w,t}(j) = \Theta_{w,t}$, $\Pi_{w,t}(j) = \Pi_{w,t}$, $c_t(j) = c_t$, $B_t(j) = B_t$, $\Lambda_t(j) = \Lambda_t$, and $m_{t,t+1}(j) = m_{t,t+1}$) gives, with $\mathrm{mrs}_t \equiv \alpha n_t^{\nu}/\Lambda_t$ and $w_t = W_t/P_t$,

$$\phi_w\big(\Pi_{w,t}-1\big)\Pi_{w,t} = (1-\eta_w)\big(1-\Theta_{w,t}\big) + \eta_w\frac{\mathrm{mrs}_t}{w_t} + \mathbb{E}_t\left[m_{t,t+1}\phi_w\big(\Pi_{w,t+1}-1\big)\Pi_{w,t+1}^2\frac{n_{t+1}}{n_t}\frac{1}{\Pi_{t+1}}\right], \tag{B.12}$$

which is (19). The first term on the right is the marginal revenue loss from a higher wage, the second the marginal saving in work effort, and the third the discounted marginal reduction in future adjustment costs. Symmetry is legitimate here because all households face the same aggregate state and the adjustment cost is convex in the household's own wage growth, so the wage distribution is degenerate at every date and no dispersion state is required.

Remark 2. Setting $\phi_w = 0$ implies $\Theta_{w,t} = 0$ and collapses (B.12) to $0 = (1-\eta_w) + \eta_w\,\mathrm{mrs}_t/w_t$, that is $w_t = \mu_w\,\mathrm{mrs}_t$ with $\mu_w = \eta_w/(\eta_w - 1)$, the constant-markup result of Erceg et al. (2000). This provides a direct numerical check on the coded model: imposing $\phi_w = 0$ should reproduce this static relationship period by period.

Remark 3. Born and Pfeifer (2020) write the household-level cost in the more general form $\frac{\phi_w}{2}\big(\Pi_{w,t}(j)/\Gamma_{t-1,t} - 1\big)^2\Xi_t$, where $\Gamma_{t-1,t}$ is an indexation term and Ξ_t a nominal scaling base. Specification (16) is the special case with no indexation and the cost measured as a share of the wage bill, which is why (B.12) contains no backward-looking term. Their slope-equivalence formula maps a given ϕ_w into the average contract duration of an observationally equivalent staggered-contract economy, which we use as an external check on the estimate of ϕ_w.

B.2 The Firm's Problem

Setup. Attaching multipliers λ_t to the flow of funds (23), q_t to the capital accumulation equation (21), φ_t to the production function (20), and μ_t to the enforcement constraint (24), the

firm's date-t Lagrangian is

$$\begin{aligned}\mathcal{L}_t = d_t + \mathbb{E}_t[m_{t,t+1}V_{t+1}(k_{t+1}, B_{t+1})] \\ + \lambda_t \left\{ y_t + \frac{B_{t+1}}{P_t R_t} - d_t - \kappa \left(d_t - \bar{d}\right)^2 - w_t n_t - I_t - \frac{B_t}{P_t} \right\} \\ + q_t \left\{ (1-\delta)k_t + I_t - \frac{\phi}{2}\left(\frac{I_t}{k_t} - \delta\right)^2 k_t - k_{t+1} \right\} \\ + \varphi_t \left\{ z_t k_t^{\theta} n_t^{1-\theta} - y_t \right\} + \mu_t \left\{ \zeta_t \left[Q_t k_t - \frac{B_{t+1}}{P_t(1+i_t)} \right] - w_t n_t \right\}. \end{aligned} \tag{B.13}$$

The firm takes w_t, i_t, P_t, $m_{t,t+1}$ and the aggregate collateral price Q_t as given. The valuation rule (26) is imposed on the symmetric equilibrium only *after* the conditions below are derived; if it were substituted into (B.13) beforehand, an individual firm would incorrectly perceive that its own investment and payout choices move the contractual price at which its collateral is assessed.

Payout. Differentiating (B.13) with respect to d_t,

$$1 - \lambda_t \left[1 + 2\kappa \left(d_t - \bar{d}\right)\right] = 0 \quad \Longrightarrow \quad \lambda_t = \frac{1}{1 + 2\kappa \left(d_t - \bar{d}\right)}. \tag{B.14}$$

The marginal value of internal funds falls below unity when the firm pays out more than its target and rises above unity when it pays out less. When $\kappa = 0$ we have $\lambda_t \equiv 1$: internal and external funds are perfect substitutes and the firm is indifferent about the timing of payouts.

Output. Since y_t appears in (23) and (20) but not in (24)—the constraint restricts the wage bill, not revenue—the credit wedge does not route through the production multiplier:

$$\lambda_t = \varphi_t. \tag{B.15}$$

This is the step that makes the labor condition below take the form of a wedge on the wage rather than on the marginal product, and it distinguishes the present specification from one in which the constraint bounds output.

labor. Differentiating with respect to n_t and using (B.15),

$$-\lambda_t w_t + \varphi_t (1-\theta)\frac{y_t}{n_t} - \mu_t w_t = 0 \quad \Longrightarrow \quad w_t \left(1 + \frac{\mu_t}{\lambda_t}\right) = (1-\theta)\frac{y_t}{n_t}, \tag{B.16}$$

which is (28). The wedge is μ_t/λ_t rather than μ_t because the shadow value of the constraint must be expressed per unit of internal funds, whose marginal value is λ_t. In the special case $\kappa = 0$ the two coincide.

Investment. Differentiating with respect to I_t,

$$-\lambda_t + q_t\left[1-\phi\left(\frac{I_t}{k_t}-\delta\right)\right]=0 \quad\Longrightarrow\quad \lambda_t = q_t\left[1-\phi\left(\frac{I_t}{k_t}-\delta\right)\right]. \tag{B.17}$$

Capital. Because (24) constrains the predetermined stock k_t rather than the choice k_{t+1}, the latter appears only in the continuation value and in (21):

$$\mathbb{E}_t[m_{t,t+1}V_{k,t+1}] - q_t = 0 \quad\Longrightarrow\quad q_t = \mathbb{E}_t[m_{t,t+1}V_{k,t+1}]\,. \tag{B.18}$$

Had the constraint been written in k_{t+1}, this condition would carry an additional contemporaneous term $\mu_t\zeta_t Q_t$, and investing today would directly relax today's borrowing limit. Under the timing adopted here the collateral benefit of capital appears instead in the envelope condition, and hence one period later in the Euler equation.

Debt. Differentiating with respect to B_{t+1}, which enters both the flow of funds and the enforcement constraint,

$$\mathbb{E}_t[m_{t,t+1}V_{B,t+1}] + \frac{\lambda_t}{P_t R_t} - \frac{\mu_t\zeta_t}{P_t(1+i_t)} = 0. \tag{B.19}$$

The second term is the marginal benefit of borrowing, discounted at the tax-advantaged rate R_t; the third is the marginal cost of consuming pledgeable collateral, since one unit of new debt reduces the right-hand side of (24) by $\zeta_t/\big(P_t(1+i_t)\big)$.

Envelope conditions. The individual state k_t enters (B.13) in three places. From the capital accumulation term, writing $u_t \equiv I_t/k_t - \delta$ and using $\partial u_t/\partial k_t = -I_t/k_t^2$,

$$\frac{\partial}{\partial k_t}\left[-\frac{\phi}{2}u_t^2 k_t\right] = -\frac{\phi}{2}\left[2u_t\left(-\frac{I_t}{k_t^2}\right)k_t + u_t^2\right] = \phi\, u_t\frac{I_t}{k_t} - \frac{\phi}{2}u_t^2. \tag{B.20}$$

From the production function, $\partial y_t/\partial k_t = \theta y_t/k_t$. From the enforcement constraint, holding the aggregate price Q_t fixed, $\partial/\partial k_t = \mu_t\zeta_t Q_t$. Collecting terms and using (B.15),

$$V_{k,t} = q_t\,\Xi_t + \lambda_t\,\theta\frac{y_t}{k_t} + \mu_t\zeta_t Q_t, \qquad \Xi_t \equiv (1-\delta) + \phi\left(\frac{I_t}{k_t}-\delta\right)\frac{I_t}{k_t} - \frac{\phi}{2}\left(\frac{I_t}{k_t}-\delta\right)^2. \tag{B.21}$$

The first two terms are the familiar user-cost and marginal-product components; the third is the collateral dividend, present only because the constraint is written in k_t. Since B_t enters only through the flow of funds,

$$V_{B,t} = -\frac{\lambda_t}{P_t}. \tag{B.22}$$

The factor $1/P_t$ is required because B_t is a nominal face value while the firm's budget is measured in goods; it is the point at which unanticipated inflation revalues the firm's liabilities.

Assembled Euler equations. Leading (B.21) one period and substituting into (B.18) gives (29). Leading (B.22) one period, substituting into (B.19) and multiplying through by P_t,

$$-\mathbb{E}_t\left[m_{t,t+1}\lambda_{t+1}\frac{P_t}{P_{t+1}}\right]+\frac{\lambda_t}{R_t}-\frac{\mu_t\zeta_t}{1+i_t}=0 \quad\Longrightarrow\quad \mathbb{E}_t\left[\frac{m_{t,t+1}\lambda_{t+1}}{\Pi_{t+1}}\right]=\frac{\lambda_t}{R_t}-\frac{\mu_t\zeta_t}{1+i_t}, \tag{B.23}$$

which is (30). Comparing (B.23) with the household's condition (B.5) is instructive and is the basis of both propositions below: households discount the same nominal payoff at $1/(1+i_t)$, while firms value it at $\lambda_t/R_t-\mu_t\zeta_t/(1+i_t)$. The tax shield drives a wedge in one direction and the collateral cost in the other, and the multiplier μ_t is precisely what reconciles them.

B.3 Collateral Valuation

Proof of Lemma 1. Dividing the valuation rule (26) by the investment condition (B.17),

$$Q_t=\frac{q_t}{\lambda_t}=\frac{q_t}{q_t\left[1-\phi\left(I_t/k_t-\delta\right)\right]}=\left[1-\phi\left(\frac{I_t}{k_t}-\delta\right)\right]^{-1}, \tag{B.24}$$

which requires only $q_t\neq 0$. Differentiating, $\partial Q_t/\partial(I_t/k_t)=\phi\left[1-\phi(I_t/k_t-\delta)\right]^{-2}=\phi\,Q_t^2$, which is strictly positive for $\phi>0$. At $I_t/k_t=\delta$ the bracket equals one, so $Q_{ss}=1$. If $\phi=0$ the bracket is identically one and $Q_t\equiv 1$. □

Two admissibility restrictions are implied and are checked numerically at every candidate parameter vector and along every simulated path:

$$\frac{I_t}{k_t}<\delta+\frac{1}{\phi} \qquad \text{and} \qquad d_t>\bar{d}-\frac{1}{2\kappa}, \tag{B.25}$$

the first ensuring a positive denominator in (40) and the second ensuring $\lambda_t>0$ in (B.14). Note also that (40) is not an additional independent equilibrium condition: it is implied by (26) together with (B.17), so imposing all three would over-determine the system.

B.4 Market Clearing and Walras' Law

Let $b_{t+1}\equiv B_{t+1}/P_t$ denote the real value at date t of the face value issued at t, so that the real repayment obligation at $t+1$ is $B_{t+1}/P_{t+1}=b_{t+1}/\Pi_{t+1}$. Written this way, the flow of funds (23) becomes

$$y_t+\frac{b_{t+1}}{R_t}=d_t+\kappa\left(d_t-\bar{d}\right)^2+w_tn_t+I_t+\frac{b_t}{\Pi_t}, \tag{B.26}$$

which makes the debt-deflation channel explicit: an unanticipated increase in Π_t reduces the real burden b_t/Π_t of inherited liabilities and thereby loosens the firm's budget.

Substituting the tax rule $T_t = b_{t+1}/R_t - b_{t+1}/(1+i_t)$ into the symmetric household budget constraint (14) cancels the two terms in $1/(1+i_t)$ and leaves

$$c_t = \big(1 - \Theta_{w,t}\big) w_t n_t + \frac{b_t}{\Pi_t} + d_t - \frac{b_{t+1}}{R_t}. \tag{B.27}$$

Substituting d_t from (B.26), the remaining debt terms and the financing subsidy cancel and we obtain (37),

$$y_t = c_t + I_t + \kappa \left(d_t - \bar{d}\right)^2 + \Theta_{w,t} w_t n_t. \tag{B.28}$$

Resources are therefore absorbed by consumption, investment, and the two adjustment costs, both of which are genuine deadweight losses. The cancellation of nominal terms is an accounting result and does not imply that the nominal block is irrelevant: inflation still enters (B.26), and the policy rate still enters both the pledgeability of collateral in (24) and the effective cost of debt in (B.23).

B.5 The Deterministic Steady State

Set all innovations to zero, $z = 1$, $\zeta = \zeta_{ss}$, and impose the inflation target $\Pi = \Pi^* = 1$. A constant real wage then implies $\Pi_w = 1$ from (31), so $\Theta_w = 0$ and the wage adjustment cost vanishes in steady state. We adopt the standard convention that the payout target $\bar{d}$ equals the steady-state payout, so $d = \bar{d}$ and, from (B.14), $\lambda = 1$. The steady state can then be solved recursively in the following order.

Step 1: rates. From (B.5) with $m = \beta$ and $\Pi = 1$,

$$1 + i_{ss} = \frac{1}{\beta}, \qquad R_{ss} = 1 + i_{ss}(1-\tau) = \frac{1 - \tau + \tau\beta}{\beta}, \qquad \beta R_{ss} = 1 - \tau + \tau\beta. \tag{B.29}$$

Step 2: capital prices. With $I = \delta k$ the investment rate equals δ, so (B.17) gives $q_{ss} = \lambda_{ss} = 1$, and (26) gives $Q_{ss} = 1$. Also $\Xi_{ss} = 1 - \delta$.

Step 3: the multiplier. Evaluating (B.23) at the steady state,

$$\beta = \frac{1}{R_{ss}} - \mu_{ss}\zeta_{ss}\beta \quad \Longrightarrow \quad \mu_{ss} = \frac{1 - \beta R_{ss}}{\zeta_{ss}\,\beta R_{ss}} = \frac{\tau(1-\beta)}{\zeta_{ss}\,(1 - \tau + \tau\beta)}, \tag{B.30}$$

using (B.29). It is convenient to define the *pledgeability-weighted wedge*

$$\Omega \equiv \mu_{ss}\zeta_{ss} = \frac{\tau(1-\beta)}{1 - \tau + \tau\beta}, \tag{B.31}$$

which depends on τ and β alone and is independent of ζ_{ss}.

Step 4: the capital-output ratio. Evaluating (29) at the steady state with $q = \lambda = Q = 1$ and

$\Xi = 1 - \delta$,

$$1 = \beta\left[(1-\delta) + \theta\frac{y}{k} + \Omega\right] \quad \Longrightarrow \quad \frac{y}{k} = \frac{1}{\theta}\left[\frac{1}{\beta} - (1-\delta) - \Omega\right]. \tag{B.32}$$

The collateral dividend Ω acts exactly like a reduction in the required marginal product of capital: because capital is useful as collateral as well as in production, the firm is willing to hold more of it than the frictionless user cost would justify, and the capital-output ratio is correspondingly higher.

Step 5: quantities and the wage. From (20), $y/k = (n/k)^{1-\theta}$, so

$$\frac{n}{k} = \left(\frac{y}{k}\right)^{\frac{1}{1-\theta}}, \qquad \frac{y}{n} = \left(\frac{y}{k}\right)^{-\frac{\theta}{1-\theta}}, \qquad w = \frac{(1-\theta)}{1+\mu_{ss}} \cdot \frac{y}{n}, \tag{B.33}$$

the last from (B.16) with $\lambda_{ss} = 1$. Normalising hours to a target $\bar{n}$ then fixes $k = \bar{n}/(n/k)$, $y = (y/k)k$, $I = \delta k$ and, from (B.28) with $\Theta_w = 0$ and $d = \bar{d}$, $c = y - \delta k$.

Step 6: debt. The enforcement constraint (24) holds with equality, so with $Q_{ss} = 1$ and $\Pi = 1$,

$$b_{ss} = (1 + i_{ss})\left[k - \frac{wn}{\zeta_{ss}}\right] = \frac{1}{\beta}\left[k - \frac{(1-\theta)y}{\zeta_{ss}\,(1+\mu_{ss})}\right]. \tag{B.34}$$

Step 7: the labor-disutility scale. With $\Pi_w = 1$ and $\Theta_w = 0$, (B.12) reduces to $w = \mu_w\,\mathrm{mrs}$, and $\Lambda_{ss} = (1 - \iota\beta)\,[(1-\iota)c]^{-\sigma}$ from (B.3). The scale parameter α is then recovered residually as $\alpha = w\,\Lambda_{ss}/\left(\mu_w \bar{n}^{\nu}\right)$, so that hours equal their target by construction.

Existence and two useful restrictions. Two conditions must hold for this steady state to be well defined, and both are imposed as constraints during estimation rather than checked only at the initial calibration. First, (B.32) requires a positive marginal product of capital,

$$\Omega = \frac{\tau(1-\beta)}{1 - \tau + \tau\beta} < \frac{1}{\beta} - (1-\delta), \tag{B.35}$$

which restricts τ relative to the steady-state real rate and depreciation. Note that this restriction falls on Ω, and hence on τ and β, and not on ζ_{ss}: by (B.31) the collateral dividend is independent of the enforcement parameter, because a lower ζ_{ss} raises the multiplier μ_{ss} in exactly offsetting proportion.

Second, (B.34) requires steady-state debt to be positive. Since $\zeta_{ss}\,(1+\mu_{ss}) = \zeta_{ss} + \Omega$, the requirement $\zeta_{ss}\,(1+\mu_{ss}) > (1-\theta)\,y/k$ is linear in ζ_{ss} and can be stated explicitly as a *lower* bound,

$$\zeta_{ss} \;>\; \underline{\zeta} \;\equiv\; (1-\theta)\frac{y}{k} - \Omega \;=\; \frac{1-\theta}{\theta}\left[\frac{1}{\beta} - (1-\delta) - \Omega\right] - \Omega, \tag{B.36}$$

where the second equality substitutes (B.32). If enforcement is too weak, the working-capital loan alone exhausts pledgeable collateral and the firm cannot support positive intertemporal debt; as $\zeta_{ss} \downarrow \underline{\zeta}$, $b_{ss} \downarrow 0$ and the model collapses to one in which firms are financed entirely by

equity and the intraperiod loan.

Remark 4. It is worth being precise about how (B.36) relates to the restriction on the enforcement parameter in Chatterjee et al. (2025), since the two are easily conflated. Setting $\Omega = 0$ in (B.36) recovers $\frac{1-\theta}{\theta}\left[\frac{1}{\beta} - (1-\delta)\right]$, which is exactly the threshold they derive. The algebraic object is therefore the same, but the inequality runs the other way and for a different reason. In their environment there is no intertemporal debt and the constraint binds only if enforcement is weak enough, so the threshold is an *upper* bound required for $\mu_{ss} > 0$. Here Proposition 1 shows that the tax advantage of debt, not the tightness of enforcement, is what makes the constraint bind: $\mu_{ss} > 0$ for *any* $\zeta_{ss} > 0$ whenever $\tau > 0$. The same threshold instead marks the point below which senior debt can no longer be supported, and so becomes a lower bound. At the calibration of Table 1 it takes the value $\underline{\zeta} \approx 0.061$.

Is there an upper bound? Neither restriction bounds ζ_{ss} from above, and no further condition does either. The natural candidate is that the steady-state payout be positive. Using (B.26) in steady state with $d = \bar{d}$ and $\Pi = 1$,

$$\bar{d} = y - wn - \delta k - b_{ss}\frac{R_{ss}-1}{R_{ss}}, \qquad \frac{R_{ss}-1}{R_{ss}} = \frac{(1-\beta)(1-\tau)}{1-\tau+\tau\beta}, \tag{B.37}$$

so the payout is what remains of output after the wage bill, depreciation and interest. As ζ_{ss} rises, μ_{ss} falls and the wage bill rises toward $(1-\theta)y$, while b_{ss} rises toward k/β; both reduce $\bar{d}$, so (B.37) is strictly decreasing in ζ_{ss}. Substituting the limiting values and using (B.32) gives

$$\lim_{\zeta_{ss}\to\infty} \bar{d} = k\left[\frac{1-\beta}{\beta} - \Omega - \frac{(1-\beta)(1-\tau)}{\beta\,(1-\tau+\tau\beta)}\right] = \frac{k(1-\beta)}{\beta\,(1-\tau+\tau\beta)}\Big[(1-\tau+\tau\beta) - \tau\beta - (1-\tau)\Big] = 0, \tag{B.38}$$

the bracket vanishing identically. The steady-state payout is therefore strictly positive for every $\zeta_{ss} > \underline{\zeta}$ and approaches zero only in the limit of perfect enforcement, in which the firm levers up until the entire operating surplus is absorbed by interest and depreciation. The admissible region is thus the half-line $\zeta_{ss} \in (\underline{\zeta}, \infty)$, and the estimation imposes only (B.35) and (B.36).

A steady-state calibration target. Equations (B.32) and (B.34) imply that the ratio of debt to quarterly output is

$$\frac{b_{ss}}{y} = \frac{1}{\beta}\left[\frac{k}{y} - \frac{1-\theta}{\zeta_{ss}\,(1+\mu_{ss})}\right], \tag{B.39}$$

which is strictly increasing in ζ_{ss} and depends on it alone once the remaining calibrated parameters are fixed. The steady state therefore maps ζ_{ss} one-to-one into observed corporate leverage, which provides a natural over-identifying check on the estimate reported in Table 1: at $\beta = 0.99$, $\tau = 0.35$, $\theta = 0.33$ and $\delta = 0.025$, equation (B.39) implies a debt-to-quarterly-output ratio of roughly six at $\zeta_{ss} = 0.15$, against a value near 0.09 that would reproduce the ratio of 3.36 targeted by Jermann and Quadrini (2012). We report this diagnostic alongside the estimates.

B.6 Proofs of the Results in Section 3.7

Proposition 1. The proof in the text is complete; (B.30) above derives the same expression as part of the recursive characterisation of the steady state. Two comparative statics are worth recording. The wedge μ_{ss} is strictly decreasing in ζ_{ss}, so a single parameter controls the average severity of the friction; and it is strictly increasing in τ, since $\partial\Omega/\partial\tau = (1-\beta)/\left(1-\tau+\tau\beta\right)^2 > 0$. The economics is that the tax shield induces firms to lever up until enforcement stops them, so a larger shield produces a tighter constraint in equilibrium, not a looser one.

Proposition 2 and Remark 1. It is convenient to prove both at once. Suppose $\kappa = 0$, so $\lambda_t \equiv 1$ by (B.14). Then (B.23) reads

$$\mathbb{E}_t\left[\frac{m_{t,t+1}}{\Pi_{t+1}}\right] = \frac{1}{R_t} - \frac{\mu_t\zeta_t}{1+i_t}. \tag{B.40}$$

The household's condition (B.5) gives $\mathbb{E}_t\left[m_{t,t+1}/\Pi_{t+1}\right] = 1/(1+i_t)$ at every date. Equating the two and multiplying by $(1+i_t)$,

$$\mu_t\zeta_t = \frac{1+i_t}{R_t} - 1 = \frac{1+i_t-\left[1+i_t(1-\tau)\right]}{R_t} = \frac{\tau i_t}{R_t} \quad\Longrightarrow\quad \mu_t = \frac{\tau i_t}{\zeta_t R_t}. \tag{B.41}$$

If in addition $\tau = 0$, then $R_t = 1+i_t$ and (B.41) gives $\mu_t\zeta_t = 0$, hence $\mu_t = 0$ whenever $\zeta_t > 0$. With $\mu_t = 0$ the enforcement constraint drops out of (B.16) and (29), and ζ_t disappears from the subsystem that determines real allocations and prices. It remains in the enforcement constraint and can therefore affect the selected debt–payout pair, but those financing choices are payoff-irrelevant when $\kappa = 0$. This proves Proposition 2.

Remark 1 is the case $\kappa = 0$ with $\tau > 0$. Expression (B.41) shows that the multiplier is then determined *statically* by the policy rate, the tax parameter and the enforcement parameter, with no reference to the collateral price Q_t, the capital stock k_t, the wage bill, or the level of debt. When $i_t > 0$, this expression implies $\mu_t > 0$ and the constraint holds with equality, determining debt compatible with the desired wage bill. When $i_t = 0$, $\mu_t = 0$ and the constraint may be slack. A negative i_t with $\tau > 0$ would imply a negative multiplier, so the unrestricted-debt, frictionless-payout conditions do not support that case with a nonnegative ICC multiplier. The result is therefore a statement about admissible equilibria, not a binding assumption. In those equilibria, collateral revaluation does not directly move the wedge. When $\kappa > 0$, λ_t is endogenous and state dependent, and the wedge depends on its expected path through (41). Collateral revaluation can then affect the wedge and the regime through the joint determination of investment, debt and payouts.

B.7 The Complete Equilibrium System

Collecting the conditions above, a symmetric equilibrium is characterised by the following nineteen conditions in the nineteen endogenous variables

$$\{c_t, X_t, \Lambda_t, n_t, w_t, \Pi_{w,t}, \Pi_t, i_t, R_t, y_t,$$
$$k_{t+1}, I_t, d_t, \lambda_t, q_t, Q_t, \mu_t, b_{t+1}, T_t\},$$

given the exogenous processes $\{z_t, \zeta_t, h_t, \chi_t, a_t\}$ and the predetermined variables $\{c_{t-1}, k_t, b_t, w_{t-1}\}$.

1. $X_t = (c_t - \iota c_{t-1})^{-\sigma}$
2. $\Lambda_t = X_t - \iota\beta\,\mathbb{E}_t X_{t+1}$
3. $1 = (1 + i_t)\,\mathbb{E}_t\left[m_{t,t+1}/\Pi_{t+1}\right]$, with $m_{t,t+1} = \beta\Lambda_{t+1}/\Lambda_t$
4. the wage Phillips curve (B.12)
5. $y_t = z_t k_t^{\theta} n_t^{1-\theta}$
6. $k_{t+1} = (1-\delta)k_t + I_t - \frac{\phi}{2}\left(I_t/k_t - \delta\right)^2 k_t$
7. $\lambda_t = \left[1 + 2\kappa\left(d_t - \bar{d}\right)\right]^{-1}$
8. $w_t\left(1 + \mu_t/\lambda_t\right) = (1-\theta)y_t/n_t$
9. $\lambda_t = q_t\left[1 - \phi\left(I_t/k_t - \delta\right)\right]$
10. $Q_t = q_t/\lambda_t$
11. the capital Euler equation (29)
12. the debt Euler equation (B.23)
13. $S_t \equiv \zeta_t[Q_t k_t - b_{t+1}/(1+i_t)] - w_t n_t \geq 0$, $\mu_t \geq 0$, $\mu_t S_t = 0$, with S_t an algebraic slack
14. the flow of funds (B.26)
15. $R_t = 1 + i_t(1-\tau)$
16. $\Pi_t = \Pi_{w,t}\, w_{t-1}/w_t$
17. the policy rule (32)
18. $T_t = b_{t+1}/R_t - b_{t+1}/(1+i_t)$
19. goods market clearing (B.28)

Four accounting relations are available—the household budget constraint, the flow of funds, the government budget, and goods market clearing—of which any three imply the fourth. The list above includes the last three, so the household budget constraint is the redundant one and is

omitted, exactly as Appendix B.4 shows. Equation (40) is likewise omitted, being implied by items 9 and 10.

Item 13 is a complementarity condition, not an equality imposed at every date. The equality is recovered in binding states; slack states have $\mu_t = 0$. The DEQN enforces this condition with (38) while allowing future regime changes to enter expectations. Proposition 1 establishes binding only at the deterministic steady state. Technology is stationary in (33), so no balanced-growth transformation is needed.

B.8 Implementation of the News Shock and Impulse Responses

Event-news state and timing. The network conditions on today's fully resolved state: any maturing event has already been confirmed or denied and its delivery incorporated into z_t. Given (χ_t, a_t), draw confirmation F_{t+1}, arrival $A_{t+1} \sim \text{Bernoulli}(\lambda_{\text{news}})$, $H_{t+1} \sim 8 + \text{Binomial}(8, p_H)$, and a new magnitude $\tilde{a}_{t+1} \sim N(0, \sigma_0^2)$ independently. Then

$$(\chi_{t+1}, a_{t+1}) = \begin{cases} (\chi_t - 1, a_t), & \chi_t > 1, \\ (H_{t+1}, \tilde{a}_{t+1}), & \chi_t \leq 1,\ A_{t+1} = 1, \\ (0, 0), & \chi_t \leq 1,\ A_{t+1} = 0. \end{cases} \tag{B.42}$$

Delivery at $t+1$ is $D_{t+1} = \mathbf{1}\{\chi_t = 1\} F_{t+1} a_t$, independent of any new announcement at that date. Thus a countdown of one resolves in the next state, not before today's decision. The same transition is used for simulation and expectations. Discrete arrival, horizon and confirmation outcomes are enumerated with their probabilities; new magnitudes are integrated by Gauss–Hermite quadrature. Two states replace the fixed-horizon relay chain used in the preliminary perturbation calculations.

The two experiments. The news intervention sets $(\chi_1, a_1) = (H, \bar{a})$ with $\bar{a} > 0$. The denied and fulfilled experiments condition respectively on $F_{1+H} = 0$ and $F_{1+H} = 1$ for this announcement. They use the same decision rule and confirmation probability during anticipation; the conditioned outcome is not revealed to agents early. Under paired background draws their paths coincide before maturity. Denial delivers no productivity increment; confirmation delivers $\bar{a}$. Without background technology innovations or additional news, the denied experiment leaves productivity unchanged at every date.

The macro-uncertainty experiment sets $\varepsilon_{h,1} > 0$ in (36) while pairing the underlying draws for enforcement and all other processes with the baseline. Enforcement draws are transformed using each path's volatility and the positive-support truncation. This preserves the implemented transition rather than artificially fixing realised enforcement.

Computing impulse responses. The nonlinear responses are generalised impulse responses. Starting states are sampled from long simulations with the DEQN. For each state, shocked and baseline paths use identical underlying random draws. Responses average their differences over both initial states and future innovations. News experiments use eligible starting states with no pending announcement and condition on the specified maturity outcome. We report simulation counts, Monte Carlo uncertainty and domain exits. Absolute multiplier and slack paths are checked separately from their responses relative to baseline, so a zero multiplier response is not mistaken for a slack constraint. No pruning is applied to the DEQN simulations.

B.9 DEQN Solution and Validation

The fixed-parameter network has nine standardised inputs, two hidden layers of 128 units each, and six outputs. Each hidden layer uses 64 tanh and 64 softplus units. The output coordinates recover the full allocation through production, the policy and tax rules, payout optimality, capital accumulation, resource clearing, and the joint financing and investment conditions. Collateral slackness is imposed through (38). These algebraic conditions hold by construction; they do not establish that the remaining equilibrium conditions have been solved.

Following Azinovic et al. (2022) and Scheidegger (2026, Chapter 2), the loss sums scaled squared residuals of the six conditions described in Section 3.6. Future allocations use the same network, evaluated at states obtained from current choices and the exogenous transitions. Expectations are weighted averages of full nonlinear expressions, not products of separate expectations. News branches are enumerated, and continuous shocks use Gauss–Hermite quadrature, including the transformation for positive enforcement. Training uses 3 volatility, 5 enforcement, 3 productivity and 3 news-magnitude nodes; validation raises each count to 7.

The network is initialised from the second-order perturbation solution. Residual scales are fixed using perturbation errors on a mixture of near-steady-state, simulated and news states. Soft level and slope anchors use the deterministic steady state and first-order solution. Adam updates the weights, with the enforcement innovation scale and stochastic volatility introduced gradually. Training states combine near-steady-state draws, persistent simulated paths, targeted news states, states near the collateral switching boundary, and a wider state-space sample. These choices follow the solution procedure in Scheidegger (2026, Chapters 2–3); they impose no desired impulse-response signs or binding dates.

Acceptance requires equation-by-equation residual checks on unseen ordinary, simulated and news states with finer quadrature, comparisons with perturbation on the same states, and checks of local slopes, feasibility and the redundant household budget. Long-horizon simulations track individual paths and report domain exits and capital and debt distributions. Binding shares are reported by state type. For the parameterised network, these checks cover interior, edge and corner parameter vectors before estimation is allowed over that region. Solver errors are checked again at the estimate using a separately trained fixed-parameter network. Small training loss or

exact accounting alone is not an acceptance criterion.

B.10 Notation

Table B.1: Notation

Symbol	Definition
Quantities	
c_t	Real consumption
n_t	Aggregate hours
y_t	Real output
I_t	Real investment
k_t	Capital stock, predetermined at t
d_t	Real equity payout; $\bar{d}$ its steady-state target
b_{t+1}	Real value of newly issued debt, B_{t+1}/P_t
B_t	Nominal face value of debt due at t
T_t	Real lump-sum tax
Prices and nominal variables	
w_t	Real wage, W_t/P_t
W_t	Aggregate nominal wage index
P_t	Price level
Π_t	Gross price inflation, P_t/P_{t-1}
$\Pi_{w,t}$	Gross nominal wage inflation, W_t/W_{t-1}
i_t	Net nominal policy rate
R_t	Firm's effective gross nominal borrowing rate, $1 + i_t(1-\tau)$
Q_t	Contractual collateral price in goods, q_t/λ_t
Multipliers and auxiliary objects	
Λ_t	Household marginal utility of wealth
X_t	Flow marginal felicity, $(c_t - \iota c_{t-1})^{-\sigma}$
$m_{t,t+1}$	Stochastic discount factor, $\beta\Lambda_{t+1}/\Lambda_t$
mrs_t	Marginal rate of substitution, $\alpha n_t^{\nu}/\Lambda_t$
λ_t	Multiplier on the flow of funds
q_t	Multiplier on capital accumulation
φ_t	Multiplier on the production function; $\varphi_t = \lambda_t$
μ_t	Multiplier on the enforcement constraint
S_t, ψ_t	ICC slack and signed DEQN collateral coordinate
Ξ_t	User-cost term defined in (B.21)

Table B.1, continued

Symbol	Definition
$\Theta_{w,t}$	Wage adjustment cost as a share of the wage bill
Ω	Steady-state pledgeability-weighted wedge, $\mu_{ss}\zeta_{ss}$
Exogenous processes	
z_t	Total factor productivity
ε_t^0, D_t	Productivity surprise and delivered news
χ_t, a_t	Remaining news horizon and pending magnitude
H, F_t	Announcement horizon and confirmation indicator
ζ_t	Enforcement parameter
h_t	Log volatility of innovations to ζ_t
$\varepsilon_{\zeta,t},\ \varepsilon_{h,t}$	Enforcement and log-volatility innovations
Parameters	
$\beta, \sigma, \iota, \nu, \alpha$	Discount factor, curvature, habit, inverse Frisch, disutility scale
η_w, μ_w	Elasticity of substitution across labor varieties; implied markup
ϕ_w	Rotemberg wage adjustment cost
θ, δ	Capital share, depreciation rate
ϕ	Investment adjustment cost
κ	Payout adjustment cost
τ	Tax advantage of debt
$\zeta_{ss}, \rho_\zeta, \bar{h}, \rho_h, \sigma_h$	Enforcement and volatility process parameters
ρ_z, σ_0	Productivity persistence and common surprise/news scale
$\lambda_{\text{news}}, p_H, \pi_{\text{confirm}}$	News arrival, horizon and confirmation parameters
ϕ_π, ϕ_y, Π^*	Policy rule coefficients and inflation target